RAMANUJAN'S PRIME COUNTING INEQUALITY HOLDS FOR ALL $x \geq 38\,358\,837\,683$

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ABSTRACT. Let N_R denote the smallest positive integer such that Ramanujan's inequality $\pi(x)^2 < \frac{ex}{\log x}\,\pi(x/e)$ holds for every real $x \geq N_R$. Assuming the Riemann hypothesis, Dudek and Platt showed that $N_R = 38\,358\,837\,683$. In this paper we prove the same value unconditionally. For this purpose we close the range $e^{103} < x < e^{3158.442}$, which was not covered by previous unconditional results, using an explicit formula for $\psi(x)$ together with new explicit information on zeros of the Riemann zeta function slightly above the height $3 \cdot 10^{12}$, up to which the Riemann hypothesis has been verified.

1. INTRODUCTION

Throughout, $\pi(x)$ is the prime counting function and $\mathrm{li}(x) = \int_0^x dt/\log t$ is the logarithmic integral (principal value). The prime number theorem in the form $\pi(x) - \mathrm{li}(x) = O(x\exp(-c\sqrt{\log x}))$, with some $c > 0$, combined with the expansion $\mathrm{li}(x) = \frac{x}{\log x}\sum_{k<n} k!\log^{-k} x + O(x\log^{-n-1} x)$, which follows by integrating by parts, determines $\pi(x)$ up to an error that is smaller than any fixed power of $\log x$ times $x/\log x$.

Ramanujan noticed that this expansion forces the inequality

$$\pi(x)^2 < \frac{ex}{\log x}\,\pi\Big(\frac{x}{e}\Big) \tag{1.1}$$

for all sufficiently large x; see Berndt's edition of the notebooks [5, p. 112]. Indeed, inserting the expansion of $\mathrm{li}(x)$ into both sides of (1.1), the terms up to order $x^2/\log^5 x$ cancel, and the right-hand side exceeds the left-hand side by $x^2/\log^6 x\,(1+o(1))$. Hence (1.1) holds as soon as the relative error $|\pi(x) - \mathrm{li}(x)|\log x/x$ is smaller than about $1/(3\log^4 x)$. As in [2], we write N_R for the least positive integer with the property that (1.1) is true for all real $x \geq N_R$.

Under the assumption that the Riemann hypothesis is true, Hassani [13] gave an explicit value of x_0 such that (1.1) holds for every $x \geq x_0$. Dudek and Platt [11] combined a refined version of this argument with a direct computation for $x \leq 1.15 \cdot 10^{16}$, and found that the largest integer counterexample to (1.1) is $x = 38\,358\,837\,682$; the largest prime counterexample, $38\,358\,837\,677$, had been found earlier by Galway (see [11]). Thus the Riemann hypothesis implies $N_R = 38\,358\,837\,683$.

Unconditionally, two ranges of x have been settled, and the history of each is short. For small x, Axler [1] showed, using Büthe's explicit bounds [7], that (1.1) holds for $38\,358\,837\,683 \leq x \leq 10^{19}$; this range was extended to $x \leq e^{58}$ by Platt and Trudgian [23] and to $x \leq e^{103}$ by Johnston [17, Theorem 5.1]. For large x, the admissible lower limit e^{X_0} has decreased from $X_0 = 9658$ (Dudek and Platt [11]) through 9032 [1], 3915 [23], 3604 [8] and 3361 [18] to $X_0 = 3158.442$ [2, Theorem 1.1]. Consequently, unconditionally,

$$38\,358\,837\,683 \leq N_R \leq e^{3158.442}.$$

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Two recent preprints of De [9, 10] claim to cover the remaining range; in Section 8 we explain why the arguments given there do not prove (1.1).

The main result of this paper closes the gap.

Theorem 1.1. *The inequality* (1.1) *holds for every real* x *with* $e^{58} \le x \le e^{3159}$. *Consequently,* $N_R = 38\,358\,837\,683$.

The second statement follows from the first one together with the results of Johnston [17, Theorem 5.1] and Axler [2, Theorem 1.1] quoted above, and the computations of Galway and of Dudek and Platt [11] for small x.

The only information about the zeros of the Riemann zeta function $\zeta(s)$ that we take from a large-scale computation is the theorem of Platt and Trudgian [24] that every non-trivial zero ρ of $\zeta(s)$ with $0 < \operatorname{Im}\rho \le H_0$, where

$$H_0 := 3 \cdot 10^{12},$$

satisfies $\operatorname{Re}\rho = 1/2$. Let us describe the method. Write $L = \log x$. By an elementary argument (Section 2), the inequality (1.1) at x follows from bounds of the form $|\pi(y) - \operatorname{li}(y)| \le \varepsilon_R(L)\, y/\log y$ at $y = x$ and $y = x/e$, where $\varepsilon_R(L)$ is an explicit function with $\varepsilon_R(L) \sim 1/(3L^4)$, and these bounds in turn follow from relative bounds for $\psi(t) - t$. To bound $\psi(t) - t$ we use a smoothed explicit formula with Logan's kernel, following Büthe [6, 7] (Section 3). The zeros of height at most H_0 lie on the critical line, while the zeros above H_0 are controlled by a zero-free region and by zero-density estimates.

The binding range turns out to be $L \approx 900$. There the kernel has bandwidth about e^{30}, and the decisive zeros are hypothetical zeros $\beta + i\gamma$ with $(1-\beta)\log\gamma \approx 0.21$ at heights just above H_0. With the published zero-free regions and zero-density estimates alone, the contribution of these zeros exceeds the available error budget several times over. The present paper therefore supplies three new explicit inputs.

(1) In Section 4 we evaluate the inequality behind the zero-free region of Bellotti, Trudgian and Yang [4] exactly at every height, instead of through a single uniform constant. This gives $(1-\beta)\log\gamma \ge 0.21384$ for zeros with $H_0 < \gamma \le e^{29}$, that is, the constant 4.676 in place of 4.896 in the zero-free region $\sigma > 1 - 1/(R\log t)$.
(2) In Section 5 we keep a second zero in the same inequality and show that two zeros close to the line $\sigma = 1$ cannot be close to each other: for example, two zeros $\beta_j + i\gamma_j$ with $H_0 < \gamma_j \le e^{29}$ and $(1-\beta_j)\log\gamma_j \le 0.348$ satisfy $|\gamma_0 - \gamma_1| \ge 0.058$.
(3) In Section 6 we rework Bellotti's log-free zero-density estimate [3] for σ close to 1. The main gain comes from an explicit and asymptotically sharp evaluation of the Barban–Vehov mean square, which is the quantity Bellotti denotes by d_4. Our bound for d_4 is about 1.5 at our parameters, while her Table 5 gives values between 85 and 105 at hers.

The proof of Theorem 1.1 is completed in Section 7 by a computation, which is carried out entirely in rigorous interval (ball) arithmetic using the Arb library [16]. Floating-point arithmetic is used only to choose parameters, and every choice yields a valid bound. The code and the output of all computations are available at `https://github.com/jaideepsaipadhi/ramanujans-inequality`. While re-deriving the explicit inputs, we found a few inaccuracies in the literature; these are collected in Section 8. None of them affects the main results of the papers concerned.

2. Preliminaries

In this section we reduce Theorem 1.1 to explicit bounds for $\psi(t) - t$. For $L > 1$ and $x = e^L$ we put

$$g(L) = \frac{\operatorname{li}(x)^2 - \frac{ex}{L}\operatorname{li}(x/e)}{(x/L)^2}, \qquad b(L) = \frac{2\operatorname{li}(x)}{x/L} + \frac{L}{L-1}, \tag{2.1}$$

and, whenever $g(L) < 0$,

$$\varepsilon_R(L) = \frac{-b(L) + \sqrt{b(L)^2 - 4g(L)}}{2}. \tag{2.2}$$

Then $\varepsilon_R(L)$ is the positive root of $\varepsilon^2 + b(L)\varepsilon + g(L) = 0$. Asymptotically, $\varepsilon_R(L) = \frac{1}{3L^4}(1 + O(L^{-1}))$; for instance, $3L^4\varepsilon_R(L)$ equals 1.2706 at $L = 58$, 1.0145 at $L = 907$ and 1.0041 at $L = 3159$.

Lemma 2.1. *Let $L = \log x \geq 58$ with $g(L) < 0$, and let $\varepsilon = \varepsilon_R(L)$. Assume that*

$$|\pi(x) - \operatorname{li}(x)| \leq \varepsilon\,\frac{x}{L} \quad \text{and} \quad \Big|\pi\Big(\frac{x}{e}\Big) - \operatorname{li}\Big(\frac{x}{e}\Big)\Big| \leq \varepsilon\,\frac{x/e}{L-1}, \tag{2.3}$$

where at least one of the two inequalities is strict. Then (1.1) holds at x.

Proof. Since $\operatorname{li}(x) > \varepsilon x/L$, the inequalities (2.3) give $0 < \pi(x) \leq \operatorname{li}(x) + \varepsilon x/L$ and $\pi(x/e) \geq \operatorname{li}(x/e) - \varepsilon x/(e(L-1))$. Hence it suffices to show that

$$\Big(\operatorname{li}(x) + \varepsilon\frac{x}{L}\Big)^2 < \frac{ex}{L}\Big(\operatorname{li}\Big(\frac{x}{e}\Big) - \varepsilon\frac{x}{e(L-1)}\Big).$$

After division by $(x/L)^2$, this inequality reads $g(L) + b(L)\varepsilon + \varepsilon^2 < 0$. For $\varepsilon = \varepsilon_R(L)$ the left-hand side vanishes, and the strict inequality in (2.3) makes the inequality strict. □

The direct evaluation of $\varepsilon_R(L)$ requires some care, since $g(L)$ is the difference of two quantities of size 1 which agree up to $O(L^{-4})$. The following lemma provides an evaluation without cancellation.

Lemma 2.2. *Let $A(L) = Le^{-L}\operatorname{li}(e^L)$, $u = 1/L$ and $P(u) = \sum_{k=0}^{15} k!\,u^k$. Then*

$$g(L) = A(L)^2 - \frac{L}{L-1}A(L-1), \qquad b(L) = 2A(L) + \frac{L}{L-1},$$

and $\varepsilon_R(L) = -2g(L)/\big(b(L) + \sqrt{b(L)^2 - 4g(L)}\big)$. Moreover, $A(L) = P(u) + R(L)$, where

$$-6e^{-L/2} \leq R(L) \leq \frac{16!\,u^{16}}{1-34u} + 6e^{-L/2} \qquad (L \geq 57). \tag{2.4}$$

In particular, $g(L) < 0$ for every $L \in [58, 3160.1]$, and on every interval $[a, a + 0.05] \subseteq [58, 3160.1]$ the function ε_R has an explicit lower bound which we compute in ball arithmetic.

Proof. The formulas for g, b and ε_R follow from (2.1) and (2.2). To prove (2.4), we integrate $\int_a^x dt/\log t$ by parts 16 times, where $a = \sqrt{x}$. Since A and P are both bounded by 1.1 for arguments at least 28.5, the boundary term at a, namely $\operatorname{li}(a) - aP(2/L)/(L/2)$, is at most $3a/(L/2)$ in absolute value. The logarithmic derivative of $e^s s^{-17}$ is at least $1 - 34/L$ on $[L/2, L]$, so the remaining integral satisfies

$$0 \leq \frac{L}{x}\,16!\int_a^x \frac{dt}{\log^{17} t} \leq \frac{16!\,u^{16}}{1-34u}.$$

This proves (2.4). Next, $S(u) = P(u)^2 - P(u/(1-u))/(1-u)$ is a power series in u with integer coefficients s_n, where $s_0 = s_1 = s_2 = s_3 = 0$, $s_4 = -1$ and $s_5 = -14$. We compute these coefficients exactly up to $n = 70$; since the coefficient of u^n in $P(u/(1-u))/(1-u)$ is $\sum_k k!\binom{n}{k} \le 2^n \sum_{k\le 15} k!$, and similarly for $P(u)^2$, the tail is at most $(\sum_{k\le 15} k!)(2u)^{71}/(1-2u)$. Writing $g(L)$ as $u^4 \cdot (S(u)/u^4)$ plus terms involving only R, we evaluate it without cancellation in ball arithmetic on each interval $[a, a+0.05]$. The enclosures for ε_R obtained in this way have relative width at most $1.9 \cdot 10^{-3}$. □

We write $y = x/e$. Then $\log y = L - 1$, and Lemma 2.1 shows that Theorem 1.1 follows from the two inequalities

$$\begin{aligned} |\pi(y) - \mathrm{li}(y)| &< \varepsilon_R(\log y + 1)\,\frac{y}{\log y} \qquad (57 \le \log y \le 3158), \\ |\pi(y) - \mathrm{li}(y)| &< \varepsilon_R(\log y)\,\frac{y}{\log y} \qquad (58 \le \log y \le 3159). \end{aligned} \tag{2.5}$$

We do not use any monotonicity of ε_R: both inequalities will be checked directly on small intervals.

To deduce (2.5) from bounds for ψ, we use Chebyshev's function $\vartheta(x) = \sum_{p\le x} \log p$ and the following identity.

Lemma 2.3. *For every $y \ge 2$, we have*

$$\pi(y) - \mathrm{li}(y) = \frac{\vartheta(y) - y}{\log y} + \int_2^y \frac{\vartheta(t) - t}{t \log^2 t}\,dt + \frac{2}{\log 2} - \mathrm{li}(2), \tag{2.6}$$

where $|2/\log 2 - \mathrm{li}(2)| < 1.8403$. Moreover, $0 \le \psi(t) - \vartheta(t) \le 1.4262\sqrt{t}$ for every $t > 0$.

Proof. By partial summation, $\pi(y) = \vartheta(y)/\log y + \int_2^y \vartheta(t)\,dt/(t\log^2 t)$, and integration by parts gives $\mathrm{li}(y) = y/\log y + \int_2^y dt/\log^2 t + \mathrm{li}(2) - 2/\log 2$. Subtracting these identities yields (2.6). The bound for $\psi(t) - \vartheta(t)$ is due to Rosser and Schoenfeld [26, Theorem 13]. □

In Section 7 we cover $[43, 3159]$ by closed intervals $[L_1, L_2]$ and show, for each of them, an inequality of the form

$$|\psi(t) - t| \le q_{[L_1,L_2]}\,\frac{\varepsilon_R^-(L_2+1)}{1 + 2/L_1}\,t + a(t) \qquad (\log t \in [L_1, L_2]), \tag{2.7}$$

where $\varepsilon_R^-(L_2+1)$ is the lower endpoint of our enclosure of $\varepsilon_R(L_2+1)$, $q_{[L_1,L_2]}$ is an explicit number, and

$$a(t) = 2 + e^{2\cdot 10^{-4}}\big(\log t + 10^{-4}\big)\Big(2.01 \cdot 10^{-4}\sqrt{t} + \frac{1}{2}\log\log(2t^2)\Big). \tag{2.8}$$

The factor $(1 + 2/L_1)^{-1}$ leaves room for the integral in (2.6).

Proposition 2.4. *Let (2.7) hold on every interval of the cover of $[43, 3159]$ constructed in Section 7, with the values $q_{[L_1,L_2]}$ obtained there. Then, for every y with $57 \le \log y \le 3159$, the bound for $|\pi(y) - \mathrm{li}(y)|$ obtained from (2.6) is at most $0.99052 \min\{\varepsilon_R(\log y + 1), \varepsilon_R(\log y)\}\, y/\log y$, where the second argument is used only for $\log y \ge 58$. In particular, (2.5) holds.*

Proof. We use Lemma 2.3, the inequality $|\vartheta(t) - t| \le |\psi(t) - t| + 1.4262\sqrt{t}$, and Büthe's bound $|\psi(t) - t| \le 0.94\sqrt{t}$ for $11 < t \le 10^{19}$ [7, Theorem 2]. For $t \le 11$ we use the trivial bound $|\vartheta(t) - t| \le 11$. Hence the part of the integral in (2.6) over $[2, e^{43}]$ is at most

$$\frac{2\,(0.94 + 1.4262)\,e^{21.5}}{\log^2 2} + 121.$$

On $[e^{43}, y]$ we substitute $t = e^v$ and bound the integral by upper sums with step 0.05 in v; on each step we use the largest bound (2.7) that meets it, together with $\int_a^b e^v v^{-2}\,dv \leq (e^b - e^a)/a^2$. For y with $\log y \in [a, a+0.05]$ and $a \geq 57$, the resulting bound for $|\pi(y) - \mathrm{li}(y)| \log y/y$ is compared with the lower bounds of Lemma 2.2 for ε_R on $[a+1, a+1.05]$ and, if $a \geq 58$, on $[a, a+0.05]$. This is carried out in ball arithmetic for every such interval $[a, a+0.05] \subseteq [57, 3159]$, and the largest ratio, 0.99052, occurs at $\log y = 877$. Note that $q_{[L_1,L_2]} < 1$ alone would not suffice, because of the additional terms $1.4262\sqrt{t}$, $a(t)$ and the integral in (2.6); the proposition rests on this computation. □

Heuristically, the factor $(1 + 2/L_1)^{-1}$ in (2.7) is explained as follows: since $\int^y \varepsilon_R(\log t + 1)\,dt/\log^2 t \approx \varepsilon_R(\log y + 1)\,\frac{y}{\log^2 y}(1 + \frac{6}{\log y})$, the integral in (2.6) adds a relative amount of about $1/L + 6/L^2 \leq 2/L$. The proof of Proposition 2.4 does not use this heuristic.

3. An explicit formula for $\psi(x)$

We use the smoothing introduced by Büthe [6, 7], but give our own proof of the smoothing step (Lemma 3.1); see Remark 8.5 for the reason. For $c > 0$ and $0 < \varepsilon \leq 10^{-4}$ let

$$\eta_{c,\varepsilon}(\tau) = \frac{c}{\varepsilon \sinh c}\, I_0\Bigl(c\sqrt{1 - (\tau/\varepsilon)^2}\Bigr)\mathbf{1}_{|\tau| \leq \varepsilon}, \qquad \ell_{c,\varepsilon}(z) = \frac{c}{\sinh c} \cdot \frac{\sinh\sqrt{c^2 - (\varepsilon z)^2}}{\sqrt{c^2 - (\varepsilon z)^2}},$$

where I_0 denotes the modified Bessel function of order 0. Then $\int \eta_{c,\varepsilon} = 2$ and $\int \eta_{c,\varepsilon}(\tau) e^{-iz\tau}\,d\tau = 2\ell_{c,\varepsilon}(z)$. Logan's function $\ell_{c,\varepsilon}$ is entire and even with $\ell_{c,\varepsilon}(0) = 1$. Let $\lambda = \int \eta_{c,\varepsilon}(\tau) e^{\tau/2}\,d\tau = 2\ell_{c,\varepsilon}(i/2) \geq 2$ and

$$\phi(\tau) = \frac{\eta_{c,\varepsilon}(\tau) e^{\tau/2}}{\lambda}, \qquad \psi_{c,\varepsilon}(x) = \int \psi_0(xe^{-\tau})\phi(\tau)\,d\tau, \tag{3.1}$$

where ψ_0 is the normalised Chebyshev function. So ϕ is a probability density on $[-\varepsilon, \varepsilon]$, and $\psi_{c,\varepsilon}(x)$ is a weighted mean of ψ_0 over $[xe^{-\varepsilon}, xe^{\varepsilon}]$; it coincides with Büthe's function [7, (3.2)–(3.4)].

Lemma 3.1. *Let $x \geq 10^{18}$, let $c \in \{30, 31, \ldots, 110\}$, let $0 < \varepsilon \leq 10^{-4}$ and put $\lambda_s = \log(10\sqrt{c}/\varepsilon)$. Assume that $\lambda_s < \log x - 1$, and let $\nu_c = \int_0^1 u\,\eta_{c,1}(u)\,du$. Then*

(3.2)

$$|\psi(x) - \psi_{c,\varepsilon}(x)| \leq \frac{\log x + \lambda_s}{\log x - \lambda_s} \cdot 2\varepsilon e^{\varepsilon/2}\Bigl(\frac{\nu_c}{2} + \frac{I_0(c)}{200\sinh c}\Bigr)x + e^{2\varepsilon}\log(e^{\varepsilon}x)\Bigl(2.01\varepsilon\sqrt{x} + \frac{1}{2}\log\log(2x^2)\Bigr).$$

Moreover, $2\sqrt{c}\,e^{\varepsilon/2}(\nu_c/2 + I_0(c)/(200\sinh c)) \leq 0.80319 < 0.7946 \cdot 1.02$ for all these c.

Proof. Since ϕ is a probability density and ψ is non-decreasing, we have

$$\psi(x) - \psi_{c,\varepsilon}(x) \leq \int_0^{\varepsilon} \bigl(\psi(x) - \psi((xe^{-\tau})^-)\bigr)\phi(\tau)\,d\tau,$$

because the integrand is non-positive for $\tau < 0$. For $3 \leq Y \leq X/2$, the Brun–Titchmarsh inequality $\pi(X) - \pi(X - Y) \leq 2Y/\log Y$ of Montgomery and Vaughan [20], together with the trivial count of prime powers as in [7, (3.9)], gives

$$\psi(X) - \psi(X - Y) \leq \log X\Bigl(\frac{2Y}{\log Y} + 0.6\frac{Y}{\sqrt{X}} + \log\log X^2\Bigr).$$

We apply this with $Y = x(1 - e^{-\tau}) \leq x\tau$. For $\tau \geq \tau_0 := \varepsilon/(10\sqrt{c})$ we have $\log Y \geq \log x - \lambda_s$, and for $\tau < \tau_0$ we bound the integrand by its value at τ_0. The mass of ϕ on $[0, \tau_0]$ is at most $\tau_0 \max\phi \leq \tau_0 e^{\varepsilon/2} c I_0(c)/(2\varepsilon\sinh c)$, and $\int_0^{\varepsilon} \tau\phi(\tau)\,d\tau \leq e^{\varepsilon/2}\varepsilon\nu_c/2$ since $\lambda \geq 2$. This gives

the upper bound for $\psi(x)-\psi_{c,\varepsilon}(x)$. The lower bound is obtained in the same way from $\psi_{c,\varepsilon}(x)-\psi(x)\le\int_{-\varepsilon}^{0}(\psi_0(xe^{|\tau|})-\psi(x))\phi(\tau)\,d\tau$; here the intervals end at xe^{ε}, which leads to the factor $\log(e^{\varepsilon}x)\le\log x+\lambda_s$. The prime powers and the term $\log\log X^2$ give the last term in (3.2). Finally, the numerical inequality for the constant is verified in ball arithmetic for each c, with ν_c computed by rigorous numerical integration; the maximum occurs at $c=110$. □

The next result is the explicit formula for $\psi_{c,\varepsilon}$ together with a bound for the contribution of zeros of large height.

Proposition 3.2 (Büthe [6, Propositions 2 and 3], [7, Proposition 2]). *Let $x\ge 10$, $0<\varepsilon\le 10^{-4}$ and $c\ge 3$. Then*

$$x-\psi_{c,\varepsilon}(x)=\frac{1}{\ell_{c,\varepsilon}(i/2)}\sum_{\rho}\ell_{c,\varepsilon}\Big(\frac{\rho-1/2}{i}\Big)\frac{x^{\rho}}{\rho}+\Theta,$$

where $|\Theta|\le 2$, and

$$\sum_{|\operatorname{Im}\rho|>c/\varepsilon}\Big|\frac{\ell_{c,\varepsilon}((\rho-\frac12)/i)}{\ell_{c,\varepsilon}(i/2)}\cdot\frac{x^{\rho}}{\rho}\Big|\le 0.16\,\frac{x+1}{\sinh c}\,e^{0.71\sqrt{c\varepsilon}}\log(3c)\log\frac{c}{\varepsilon}.$$

Lemma 3.3. *Let $\rho=\beta+i\gamma$ with $0<\beta<1$ and $|\varepsilon\gamma|\le c$. Then $|\ell_{c,\varepsilon}((\rho-1/2)/i)|\le e^{\varepsilon(c+1)/6}\ell_{c,\varepsilon}(\gamma)$ and $\ell_{c,\varepsilon}(i/2)\ge 1$.*

Proof. We write $\ell_{c,\varepsilon}(z)=\frac{c}{\sinh c}G(c^2-\varepsilon^2z^2)$ with $G(w)=\sum_{n\ge0}w^n/(2n+1)!$. For $z=\gamma-i(\beta-\frac12)$ we get $|c^2-\varepsilon^2z^2|\le c^2-\varepsilon^2\gamma^2+\varepsilon(c+1)$. Since G has positive coefficients, $|G(w)|\le G(|w|)$, and since $\frac{d}{dw}\log G(w)\le 1/6$ for $w\ge0$, the first claim follows. The second claim holds because G is increasing on $[0,\infty)$. □

Lemma 3.4. *For every $T\ge 10^4$, we have $\sum_{0<\gamma\le T}1/\gamma\le\frac12\log^2T$, where the sum runs over the zeros of $\zeta(s)$ counted with multiplicity.*

Proof. Let $N(T)$ denote the number of zeros with $0<\gamma\le T$. We have $\sum_{0<\gamma\le T}1/\gamma=N(T)/T+\int_{14}^{T}N(t)t^{-2}\,dt$. Inserting the bound $|N(t)-\frac{t}{2\pi}\log\frac{t}{2\pi e}-\frac78|\le 0.1038\log t+0.2573\log\log t+9.3675$ of Hasanalizade, Shen and Wong [12], we obtain $\sum_{0<\gamma\le T}1/\gamma\le\frac{1}{4\pi}\log^2T+\log T+2$, which is smaller than $\frac12\log^2T$ for $T\ge10^4$. □

From now on let $H=H_0=3\cdot10^{12}$ and $T_1=c/\varepsilon$, and for $s\in[0.6,1)$ let

$$S(s)=e^{\varepsilon(c+1)/6}\sum_{\substack{H<\gamma\le T_1\\ \beta>s}}\frac{\ell_{c,\varepsilon}(\gamma)}{\gamma}.$$

Further, let $N(\sigma,T)$ denote the number of zeros $\beta+i\gamma$ of $\zeta(s)$ with $\beta>\sigma$ and $0<\gamma\le T$, counted with multiplicity. The following proposition is the basis of the computation in Section 7.

Proposition 3.5. *Let $43\le L_1<L_2$, $c\in\{30,\dots,110\}$ and $0<\varepsilon\le10^{-4}$ with $\lambda_s=\log(10\sqrt{c}/\varepsilon)<L_1-1$. Assume that every zero $\beta+i\gamma$ with $H<\gamma\le T_1$ satisfies $\beta\le s_{\max}$. Then, for every x with $\log x\in[L_1,L_2]$,*

$$\begin{aligned}\frac{|\psi(x)-x|}{x}\le\frac{L_2+\lambda_s}{L_1-\lambda_s}\cdot\frac{0.7946\cdot1.02\,\varepsilon}{\sqrt{c}}+\frac{0.32\,e^{0.71\sqrt{c\varepsilon}}\log(3c)\log T_1}{\sinh c}+e^{-L_1/2}\log^2H\\ +2\Big(L_2\int_{0.6}^{s_{\max}}e^{(s-1)L_1}S(s)\,ds+e^{-0.4L_1}\big(S(0.6)+\log^2T_1\big)\Big)+\frac{a(x)}{x},\end{aligned}\tag{3.3}$$

where $a(x)$ is given by (2.8).

Proof. We combine Lemma 3.1 with Proposition 3.2. Since $0.16(x+1) \leq 0.32x$, the zeros with $|\gamma| > T_1$ contribute the second term. The zeros with $|\gamma| \leq H$ lie on the critical line [24], so by Lemmas 3.3 and 3.4 they contribute at most $2x^{1/2} \cdot \frac{1}{2}\log^2 H$. The zeros with $H < |\gamma| \leq T_1$ occur in pairs $\rho, \bar{\rho}$, which gives the factor 2. For those with $\beta > 0.6$ we use the identity

$$e^{\varepsilon(c+1)/6} \sum_{\substack{H<\gamma\leq T_1 \\ \beta>0.6}} x^{\beta-1} \frac{\ell_{c,\varepsilon}(\gamma)}{\gamma} = x^{-0.4} S(0.6) + \log x \int_{0.6}^{1} x^{s-1} S(s)\, ds,$$

together with Lemma 3.3, the inequalities $x^{s-1} \leq e^{(s-1)L_1}$ and $\log x \leq L_2$, and $S(s) = 0$ for $s \geq s_{\max}$. All remaining zeros in this range of heights have $\beta \leq 0.6$; they contribute at most $x^{-0.4} e^{\varepsilon(c+1)/6} \sum_{H<\gamma\leq T_1} 1/\gamma \leq x^{-0.4} \log^2 T_1$, since Lemma 3.4 gives $\frac{1}{2}\log^2 T_1$ and $e^{\varepsilon(c+1)/6} \leq e^{0.002} < 2$. The terms Θ and those of Lemma 3.1 give $a(x)/x$. □

To bound $S(s)$ we use summation by parts.

Lemma 3.6. *Let $f(t) = \ell_{c,\varepsilon}(t)/t$ for $\varepsilon t < c$ and $f(t) = c/(t \sinh c)$ for $\varepsilon t \geq c$, and let $H = t_0 < t_1 < \cdots < t_n$ with $t_n \geq T_1$. Let $N^+(s,t)$ be any upper bound for $N(s,t)$, and assume that $\lambda^*(l)/l$ is non-increasing, where every zero with $\gamma > H$ satisfies $(1-\beta)\log\gamma \geq \lambda^*(\log\gamma)$. If we replace $N^+(s,t_{i+1})$ by 0 whenever $s \geq 1 - \lambda^*(\log t_{i+1})/\log t_{i+1}$, then*

$$S(s) e^{-\varepsilon(c+1)/6} \leq \sum_{i=0}^{n-1} N^+(s,t_{i+1})\big(f(t_i) - f(t_{i+1})\big) + N^+(s,t_n) f(t_n).$$

Proof. The function f is positive and non-increasing on $(0,\infty)$; enlarging the range of heights from T_1 to t_n can only increase the sum. Zeros with $\gamma \leq H$ do not satisfy $\beta > s \geq 0.6$. Hence the inequality follows by summation by parts. Since $\lambda^*(l)/l$ is non-increasing, $s \geq 1 - \lambda^*(\log t_{i+1})/\log t_{i+1}$ implies $N(s,t_{i+1}) = 0$. □

4. A zero-free region depending on the height

Bellotti, Trudgian and Yang [4, Theorem 1] proved that $\zeta(\sigma + it) \neq 0$ for $\sigma > 1 - 1/(4.896\log t)$ and $t \geq 3$. The constant 4.896 is determined by heights near e^{76}. At the heights relevant to us, namely $\log t \leq 43$, their basic inequality permits a smaller constant. In this section we evaluate it exactly at every height. Since several constants in [4] are computed only for $\sigma \geq 0.99346$, we first re-derive them for the smaller values of σ and the larger values of η that we need.

4.1. The inequality of Bellotti, Trudgian and Yang.

We keep the notation of [4]. Let $\theta = 1.1338$, and let w be the function of Heath-Brown type defined in [4, Definition 1], which is supported on $[0,D]$ with $D = 2\theta\cot\theta$. Let $W(z) = \int_0^D e^{-zu} w(u)\, du$ and $w_0 = w(0) = 5.672787598\ldots$. Let κ_m $(0 \leq m \leq 6)$ be the coefficients in [4, (18)], with $\kappa = \sum_m \kappa_m = 433/859$, and let $P(x) = \sum_{k\leq 16} a_k \cos(kx) \geq 0$ be the trigonometric polynomial of [4, Section 4], with $a_0 = 1$, $a_1 = 865534/497079$ and $a = \sum_{k\geq 1} a_k = 2919857/828465$. Finally, let $K = 16$ and $T_0 = 10^{10}$. For $\eta > 0$ and $\frac{1}{2} < \sigma < 1$ we put

$$f(u) = \eta w(\eta u) \sum_m \kappa_m e^{-(2\sigma-1)mu}, \qquad F(z) = \sum_m \kappa_m W\Big(\frac{z + (2\sigma-1)m}{\eta}\Big),$$

so that $f(0) = \kappa\eta w_0$.

Let $\rho_0 = \beta_0 + it$ be a zero with $t \geq H_0$, and put $\eta = 1-\beta_0$ and $\lambda_0 = \eta \log t$. Suppose that $\zeta(s)$ has no zeros $\beta+i\gamma$ with $\beta \geq 1-A/\log\gamma$ and $H_0 \leq \gamma \leq Kt+T_0$, and put $\sigma = 1-A/\log(Kt+T_0)$. Assume that $\eta \leq \eta_0 \leq 0.0145$ and $\sigma \geq \sigma_0 \geq 0.993$. Then the argument of [4, Lemma 15] gives

$$c_{10}\,\eta w_0 \leq \eta w_0\Big(\frac{a\kappa}{2}\log t - G\Big) + a_0 F(\sigma-1) - a_1\big(F(\sigma-1+\eta)+F(\sigma-\eta)\big) - Z + 10^{-10}\eta + 2\cdot 10^{-7} + E_2\lambda_0\eta^2 + E_3\eta^3, \tag{4.1}$$

where $Z \geq 0$ is the part of the sum over zeros that is discarded, and the constants c_{10}, G, E_2, E_3 depend on (η_0, σ_0) and are given in Lemmas 4.3–4.5 below. Consequently, there is no such zero as soon as

$$\mathcal{M}_1 := a_1\big(F(\sigma-1+\eta)+F(\sigma-\eta)\big) - \Big[\eta w_0\Big(\frac{a\kappa}{2}\log t - G - c_{10}\Big) + a_0F(\sigma-1) + 10^{-10}\eta + 2\cdot 10^{-7} + E_2\lambda_0\eta^2 + E_3\eta^3\Big] \tag{4.2}$$

is positive.

For the convenience of the reader we recall how (4.1) arises, so that it is clear which parts of [4] are used without change. Put $s_k = \sigma + ikt$ for $0 \leq k \leq K$. The starting point is the identity of [4, Lemma 6]:

$$\sum_n \frac{\Lambda(n)}{n^\sigma} f(\log n) P(t\log n) = \sum_{k=0}^{K} a_k \operatorname{Re}\Big(F(s_k-1) - \sum_\rho F(s_k-\rho) + \Gamma_k + E_k\Big),$$

where Γ_k is the contribution of the archimedean factor and E_k is the error made in replacing f by its main part, together with tail terms. The left-hand side is bounded from below by $c_{10}\eta w_0$ in Lemma 4.4. On the right, the pole term with $k=0$ is $a_0F(\sigma-1)$; the pole terms with $k \geq 1$ are at most $10^{-10}\eta$ in total by [4, Lemma 13]. In the sum over zeros with $k=1$ we keep ρ_0 and $1-\bar\rho_0$, which give $-a_1(F(\sigma-1+\eta)+F(\sigma-\eta))$; all other zeros are grouped in pairs ρ, $1-\bar\rho$, each pair contributes a non-positive amount by Lemma 4.2, and Z denotes minus their total. The archimedean terms are bounded by Lemma 4.5, the terms E_k by Lemma 4.3, and the remaining tails of [4, Lemma 6] by $2\cdot 10^{-7}$. Thus, from [4] we use the definitions of w, κ_m and P, the identity of [4, Lemma 6] and the bound of [4, Lemma 13]; all constants that depend on (η_0, σ_0) are re-derived below and evaluated in ball arithmetic. In [4], the term coming from their Lemma 10 enters the final inequality with $f(0) = \eta w_0$ instead of $\kappa\eta w_0$; see Remark 8.1.

4.2. **The constants.**

Lemma 4.1. *Let $W_0(z) = W(z) - w_0/z$, and let $\nu, r > 0$. For $\operatorname{Re} z \geq \nu$ and $|z| \geq r$ we have $|W_0(z)| \leq C_w(\nu,r)|z|^{-3}$, where*

$$C_w(\nu,r) = |w''(0)| + \frac{1}{r}\Big(|w'''(0)| + e^{-\nu D}|w'''(D)| + \int_0^D |w''''(u)|e^{-\nu u}\,du\Big),$$

$|w''(0)| = 50.06970149\ldots$ and $|w'''(0)| = |w'''(D)| = 117.26477652\ldots$.

Proof. We have $w(D) = w'(D) = w''(D) = 0$ and $w'(0) = 0$. Hence four integrations by parts give

$$W(z) = \frac{w_0}{z} + \frac{w''(0)}{z^3} + \frac{w'''(0) - e^{-zD}w'''(D)}{z^4} + \frac{1}{z^4}\int_0^D e^{-zu}w''''(u)\,du,$$

and the lemma follows. The values of the derivatives are enclosed in ball arithmetic; the integral is decreasing in ν, so it suffices to bound it at integers $\nu \in [30, 200]$. □

For example, $C_w(69.4, 69.4) = 51.80$, while [4, Lemma 4] uses the bound 52 at this point.

Lemma 4.2. *Let $\sigma \in [0.993, 1)$ and $\eta \leq \eta_0 \leq 0.0145$. Then $\operatorname{Re} F(z) + \operatorname{Re} F(2\sigma - 1 - \bar{z}) \geq 0$ for $0 \leq \operatorname{Re} z \leq 2\sigma - 1$.*

Proof. As in [4, Lemma 5], it suffices to consider $\operatorname{Re} z = 0$. For fixed σ, the shifts in F are exactly $(2\sigma - 1)m$, and the substitution $y \mapsto y/(2\sigma - 1)$ reduces the claim to

$$B(y) := \sum_{m=1}^{7} b_m \frac{m - \operatorname{sgn}(b_m)\varepsilon_0/m}{m^2 + y^2} > 0 \quad (y \in \mathbb{R}), \qquad \varepsilon_0 = \frac{C_w \eta_0^2}{w_0(2\sigma - 1)^2},$$

where $(1 + x)\sum_m \kappa_m x^m = \sum_m b_m x^m$. For $\eta_0 \leq 0.0145$ and $\sigma \geq 0.993$ we have $\varepsilon_0 \leq 1/500$, using $C_w \leq 51.9$. For $\varepsilon_0 = 1/500$, the function B is rational, its value at 0 is positive, and its numerator has no real zeros, as we verify by exact root isolation. Since B decreases as ε_0 increases, the claim follows. □

Lemma 4.3. *Let $x_0 = \sigma_0 - \frac{1}{2}$, $C_1 = C_w(x_0/\eta_0, x_0/\eta_0)$, $C_2 = C_w(\sigma_0/\eta_0, \sigma_0/\eta_0)$ and $\Sigma_\kappa = \sum_m |\kappa_m|$. Then the error terms E_k of* [4, Lemma 8] *satisfy $\sum_k a_k |E_k| \leq E_2 \lambda_0 \eta^2 + E_3 \eta^3$, where*

$$E_2 = a\Sigma_\kappa \frac{C_1}{\pi x_0^2}, \qquad E_3 = a_0 \Sigma_\kappa \Big(\frac{23.6497\, C_1}{2\pi} + \frac{C_2}{x_0^3} \Big) + a\Sigma_\kappa \frac{C_2}{x_0^3}.$$

Proof. We follow the proof of [4, Lemma 8] with two changes. First, the inequality $\eta \log t \leq A_0$ used in [4, (33)] is replaced by the identity $\eta \log t = \lambda_0$. Second, the integral [4, (32)] is evaluated with the digamma function itself: we have $\int_{\mathbb{R}} |\operatorname{Re} \frac{\Gamma'}{\Gamma}(\frac{1}{4} + \frac{iy}{2})| (x_0^2 + y^2)^{-3/2}\, dy \leq 23.6497$ for $x_0 \geq 0.493$, since the integral is decreasing in x_0. □

Lemma 4.4. (i) *$f(u) \geq 0$ for $u \geq 0$.*

(ii) *$f(u) \geq \kappa f(0)$ for $0 \leq u \leq u_0(\eta_0, \sigma_0)$; for example, $u_0 = 59.15$, 49.46 and 33.61 for $\eta_0 = 0.0071$, 0.0085 and 0.0125, respectively.*

(iii) *$\sum_n \Lambda(n) n^{-\sigma} f(\log n) P(t \log n) \geq c_{10} \eta w_0$, where $c_{10} = \kappa^2 \sum_{15 \log p \leq u_0} m_p \log p$, and m_p are the lower bounds for $\min_x G_p(1, x)$ in* [4, Table 1]*, except that $m_5 = 0.0073564$.*

Proof. (i) The function w is non-negative on $[0, D]$; we verify this by bisection in ball arithmetic on $[0, D - 0.02]$ and by a Taylor bound near D, where $w'''(D) < 0$. The polynomial $\sum_m \kappa_m x^m$ has no zero in $[0, 1]$.

(ii) We verify $w(\eta_0 u) \sum_m \kappa_m e^{-d_m u} - \kappa w_0 \geq 0$, with d_m as in [4, Lemma 11], by bisection in ball arithmetic; near $u = 0$ we use that this function vanishes at 0 and has derivative at least 1.19 there.

(iii) As in [4, Lemma 10], the terms with $n > e^{u_0}$, the primes with $15 \log p > u_0$ and the prime powers p^m with $m > 15$ may be omitted by (i) and $P \geq 0$. The inequality $G_p(\sigma, x) \geq G_p(1, x)$ holds because $G_p(\sigma, x) = \sum_{m \leq 15} p^{-m\sigma} P(mx)$ is non-increasing in σ. The values m_p are verified by bisection with a mean value enclosure; see Remark 8.2 for $p = 5$. □

Lemma 4.5. *In the notation of* [4, Lemma 12]*, we have*

$$\sum_k a_k \sum_m \kappa_m \Big(-\frac{1}{2} \log \pi + \frac{1}{2} \operatorname{Re} \frac{\Gamma'}{\Gamma} \Big(\frac{s_k + \delta_m}{2} + 1 \Big) \Big) \leq \frac{a\kappa}{2} \log t - G(\sigma_0),$$

where $G(\sigma_0)$ is given by the formula in [4]*; in particular, $G(0.993) = 1.56839$ and $G(0.9935164) = 1.56888$.*

Finally, the term $10^{-10}\eta$ of [4, Lemma 13] and the tail estimates in [4, Lemma 6] remain valid for $\eta \leq 0.015$ and $\log t \leq 45$; each tail term is at most $3 \cdot 10^{-9}$, so their sum is less than $2 \cdot 10^{-7}$.

4.3. **Evaluation of W.** On $[0,D]$, the function w is a linear combination of $u\cos(ku)$, $\cos(ku)$, $\sin(\phi-ku)$, u and 1, where $k=\tan\theta$. Hence W has a closed form. With $J_0(s)=(1-e^{-sD})/s$, $J_1(s)=(1-e^{-sD}(1+sD))/s^2$ and $\Sigma_\phi(z)=(e^{i\phi}J_0(z+ik)-e^{-i\phi}J_0(z-ik))/(2i)$, it reads

$$W(z)=\sec^2\theta\Big[\frac{\sec^2\theta}{2}\sum_{\pm}\Big(\theta\cot\theta\,J_0-\frac{1}{2}J_1\Big)(z\mp ik) \\ +(2\theta\cot\theta-2)J_0(z)-J_1(z)+\frac{\Sigma_{2\theta}(z)}{\sin 2\theta}-\frac{2\Sigma_\theta(z)}{\sin\theta}\Big].$$

In the verification we evaluate W on complex balls with the second-order enclosure

$$W(z)\in W(z_c)+W'(z_c)(z-z_c)+B\Big(0,\frac{r^2}{2}\sup|W''|\Big),\qquad r=|z-z_c|,$$

where W' has the same closed form with J_0,J_1 replaced by $-J_1,-J_2$. Three integrations by parts of $u^2w(u)$ give $\sup|W''|\le\min\{\int_0^D u^2we^{-\nu u},\ (2w_0+\int_0^D|(u^2w)'''|e^{-\nu u})|z|^{-3}\}$.

4.4. **The zero-free region.**

Theorem 4.6. *Let λ^* be the step function defined on the grid $l\in\{28,28.25,\dots,43\}$ by the values computed in Section 7, part of which are listed below, and extended by $\lambda^*(l)=\lambda^*(l')$ for $l\le 43$, where l' is the smallest grid point with $l'\ge l$, and by $\lambda^*(l)=1/4.896$ for $l>43$. Then every zero $\beta+i\gamma$ of $\zeta(s)$ with $\gamma>H_0$ satisfies*

$$(1-\beta)\log\gamma\ge\lambda^*(\log\gamma).$$

l	28	29	30	31	32	33
$\lambda^*(l)$	0.21424	0.21384	0.21345	0.21308	0.21272	0.21239
l	34	35	36	37	38	40
$\lambda^*(l)$	0.21206	0.21175	0.21145	0.21117	0.21084	0.21033

Moreover, $\lambda^*(l)/l$ is non-increasing.

Proof. For $\log\gamma>43$ the claim is [4, Theorem 1]. For $\log\gamma\le 43$ we use an iteration with four steps, starting from $\lambda_0=1/4.896$. At step j we assume the conclusion of step $j-1$. Since $\lambda_{j-1}(l)/l$ is non-increasing, this assumption provides the zero-free input in (4.1) with $A=\lambda_{j-1}(\log(Kt+T_0))$. In every interval of heights $\log t\in(l-0.25,l]$ we show $\mathcal{M}_1>0$ in ball arithmetic, for all $\log t$ in the interval and all $\lambda_0=\eta\log t\in[\lambda_{j-1}(l),\lambda_j(l)]$, by adaptive bisection. On each box, σ is taken as a ball with A at the top of the box, which is admissible since a smaller value of A is a weaker hypothesis, and the constants are taken at $\eta_0=\lceil\lambda_j(l)/l_{\min}\rceil_{10^{-4}}+10^{-4}$ and at σ_0 rounded down. Then a zero with $\lambda_{j-1}(l)\le(1-\beta)\log\gamma<\lambda_j(l)$ would satisfy (4.1) with $\mathcal{M}_1>0$, which is impossible. The table of the fourth step differs from that of the third by at most 10^{-5} in each entry. □

For example, at $\log t=29$ Theorem 4.6 gives the zero-free region $\sigma>1-1/(4.676\log t)$.

5. Pairs of zeros close to the line $\sigma=1$

In [4, Lemma 6], the sum over zeros $\mathrm{Re}\sum_\rho F(s_1-\rho)$ is bounded from below by the contribution of the pair ρ_0, $1-\bar\rho_0$, while every other zero ρ is paired with $1-\bar\rho$, and every such pair contributes a non-negative amount by Lemma 4.2. If there is a second zero close to ρ_0, we may keep its pair as well.

Lemma 5.1. *Assume the situation of* (4.1)*, and let* $\rho_1 = \beta_1 + i(t+d)$ *be a second zero with* $\beta_1 > \frac{1}{2}$ *and* $\sigma - \beta_1 > 0$*. Let* $\eta_1 = 1 - \beta_1$*. Then* (4.1) *holds with*

$$Z \geq a_1 \operatorname{Re}\big(F(\sigma - 1 + \eta_1 - id) + F(\sigma - \eta_1 - id)\big).$$

Proof. The pair ρ_1, $1 - \bar{\rho}_1$ contributes exactly the displayed quantity, since the point s_1 enters with the coefficient a_1. All other pairs contribute non-negative amounts by Lemma 4.2, since $0 \leq \operatorname{Re}(s_1 - \rho) \leq 2\sigma - 1$ for every zero below height $Kt + T_0$. Since $F(\bar{z}) = \overline{F(z)}$, only $|d|$ matters. □

Consequently, two zeros at heights t and $t + d$ cannot exist if

$$\mathcal{M}_2 := \mathcal{M}_1 + a_1 \operatorname{Re}\big(F(\sigma - 1 + \eta_1 - id) + F(\sigma - \eta_1 - id)\big) > 0$$

for one of the two choices of the primary zero, where F is formed with the η of the primary zero. Heuristically, $\operatorname{Re} F(\sigma - \eta_1 - id) \approx \eta w_0(\sigma - \eta_1)/((\sigma - \eta_1)^2 + d^2)$, which is about 0.04 for $|d| \leq 0.05$ and $\eta, \eta_1 \approx 0.0075$; this term causes the repulsion.

Theorem 5.2. *Let* $29 \leq l_k \leq 37$ *be an integer, and let* k *be an integer with* $\lambda^*(l_k) < k \cdot 10^{-4} l_k \leq 0.0125(l_k - 1)$*. There are explicit numbers* $d^*(l_k, k) \geq 0$ *with the following property. If two zeros* $\beta_j + i\gamma_j$ $(j = 0, 1)$ *satisfy* $H_0 < \gamma_j \leq e^{l_k}$ *and* $1 - \beta_j \leq k \cdot 10^{-4}$*, then* $|\gamma_0 - \gamma_1| \geq d^*(l_k, k)$*. For each* l_k*, the value* $d^*(l_k, k)$ *is largest for the smallest admissible* k *(for instance* 0.142 *for* $l_k = 29$*), and for large* k *it is at least* 0.058 *for* $l_k = 29$*,* 0.056 *for* $l_k = 30$ *and* 0.044 *for* $l_k = 37$*. Outside the stated range of* (l_k, k) *we put* $d^*(l_k, k) = 0$*.*

Proof. Let $e_{\min} = \lambda^*(l_k)/l_k$ and $e_{\max} = k \cdot 10^{-4}$. By Theorem 4.6 and the monotonicity of $\lambda^*(l)/l$, both zeros satisfy $1 - \beta_j \geq e_{\min}$; since $e_{\min} > A/\log(Kt + T_0)$ (for example $0.00737 > 0.0067$ for $l_k = 29$), the hypothesis $\sigma - \beta_1 > 0$ of Lemma 5.1 is satisfied. Fix an interval of heights $(l - 1, l]$ with $l \leq l_k$. We show $\mathcal{M}_2 > 0$ for all $(\log t, \eta_0, \eta_1, |d|) \in [l - 1, l + 10^{-12}] \times [e_{\min}, e_{\max}]^2 \times [0, d^*]$ in ball arithmetic by adaptive bisection, trying both orderings of the pair on each box; the extra width 10^{-12} accounts for the height of the second zero. As in the proof of Theorem 4.6, σ is a ball with A at the top of the box, and the constants are taken at outward rounded values of (η_0, σ_0). We do this for the values of k on a coarse grid, with d^* chosen slightly below the largest value suggested by a floating-point computation; for the remaining k we use the value of the next larger k on the grid, which is admissible since enlarging $e_{\max}$ only enlarges the box. Since a zero with height at most e^{l_k} may lie in any lower interval of heights, we finally replace $d^*(l_k, k)$ by $\min_{l \leq l_k} d^*(l, k)$, where the minimum is taken over those l for which $1 - \beta \leq k \cdot 10^{-4}$ is compatible with Theorem 4.6. □

6. Zero-density estimates

In the computation, $N(s, t)$ is replaced by an upper bound $N^+(s, t)$, which is the minimum of the following bounds, each in its range of validity.

- (N1) $N(0, T) \leq \frac{T}{2\pi} \log \frac{T}{2\pi e} + \log T + 1$ for $T \geq H_0$, which follows from [12].
- (N2) The bounds of Kadiri, Lumley and Ng [19, Table 1], in the form $N(\sigma, T) \leq A(\log T)^{5 - 2\sigma_0} T^{8(1-\sigma_0)/3} + B \log^2 T$ for $\sigma \geq \sigma_0 \in \{0.60, 0.65, \ldots, 0.99\}$ and $T \geq 3.061 \cdot 10^{10}$.
- (N3) The bounds $N(\sigma, T) \leq C T^{B(1-\sigma)}$ of Bellotti [3, Theorem 1.1 and Table 2] for $\sigma \geq 0.9927$ and $T > H_0$, both the uniform one and those for $e^{l-1} < T \leq e^l$ with $29 \leq l \leq 41$.
- (N4) Theorem 6.1 below, for $1 - \sigma \leq 0.03$ and $28 < \log T \leq 40$.

In the binding range only (N4) matters. In this section we prove Theorem 6.1 by revisiting Bellotti's log-free zero-density estimate [3, Sections 3–4].

6.1. **The log-free estimate.** In this section $L = \log T$, and we write $U = T^u$, $V = T^v$ with $v = \tau u$, $W = T^w$ and $X = T^x$. Let ξ_d be the Barban–Vehov weights, supported on $d \le V$, and $\Psi(n) = \sum_{d|n} \xi_d$, so that $\Psi(1) = 1$ and $\Psi(n) = 0$ for $1 < n \le U$. Let θ_d be the Selberg-type weights, supported on $d \le W$, and $\Theta(n) = \sum_{d|n} \theta_d$. Let $\alpha = 1 - \delta$. We divide the strip into rectangles of height $h = \kappa_0 \delta$, choose in each rectangle one zero with $\beta \ge \alpha$ as its representative if there is one, and let $\tilde{n}$ be an upper bound for the number of zeros with $\beta \ge \alpha$ in a rectangle.

Theorem 6.1. *Let $k \le 300$ be a positive integer, $\delta_k = k \cdot 10^{-4}$ and $\delta \in ((k-1)10^{-4}, \delta_k]$. Let $l \in \{29, 29.5, 30, 30.5, 31, 32, \dots, 40\}$ and $\log T \le l$. Then, for every choice of parameters (u, τ, w, x) which is admissible in the sense of Lemma 6.2,*

$$N(1-\delta, T) \le N(1-\delta_k, e^l) \le \tilde{n}\, \frac{d_4 d_5}{w} \cdot \frac{\mathcal{C}}{c_1^2 - c_5} \cdot \frac{e^{2x\delta_k l}}{\delta_k l},$$

where the constants $\tilde{n}$, d_4, d_5, $\mathcal{C}$, c_1 and c_5 are given in Lemma 6.2 with $\delta = \delta_k$ and $L = l$.

The first inequality in Theorem 6.1 is the monotonicity of $N(\sigma, T)$; in particular, the bound is applied at the corner of each cell and no rescaling inside a cell is used. For each cell (k, l) we choose the parameters numerically so that the bound is small; since every admissible choice gives a valid bound, this optimisation needs no verification.

Lemma 6.2. *Let $\alpha_k = 1 - \delta_k$ and $\nu = \lambda^*(l)/l$. We call (u, τ, w, x) admissible if $\delta_k > \nu$ and*

$$0.02 < w < u, \qquad 1.3 \le \tau \le 2.5, \qquad uL \ge 40,$$
$$uL - 2\log L < xL, \qquad 1.1\,\tau u \le x \le 2.6\,\tau u, \qquad 2\nu x L \ge 1,$$

and if the constants c_1 and c_5 below satisfy $c_1 > 0$ and $c_1^2 - c_5 \ge 0.2$. For admissible parameters, Theorem 6.1 holds with the following constants.

(i) $c_1 = e^{-L^2/U} - B_1 - 3.8 e^{-L(L-2u)} \dfrac{L^2+1}{w(v-u)L^6}$, *where*

$$B_1 = \frac{10.45 \cdot 0.618 \cdot 5.922\, L}{2\pi}\, T^{(v+w)/2 + 1/6 + x(1/2 - \alpha_k)}.$$

(ii) $d_5 = 1.0061\big(1 + (1.333 + 4e^{-wL/2})/(wL)\big)$ [3, Lemma 4.3].

(iii) d_4 *is any upper bound for* $\sum_n \Psi(n)^2 n^{-1}(e^{-n/X} - e^{-nL^2/U})$; *we use Theorem 6.5.*

(iv) $c_5 = \dfrac{d_4 d_5}{2\pi} \cdot \dfrac{e^{wL}}{wL}\Big(e^{Lx(3/2 - 2\alpha_k)} + e^{(Lu - 2\log L)(3/2 - 2\alpha_k)}\Big) \cdot 4.72 \cdot 0.618\, e^{L/6} L.$

(v) $\mathcal{C} = 1.0067\big(1 + 2\delta_k \mathcal{L}(\kappa_0 \delta_k)\big) + 2\sqrt{2\pi} \cdot 2.4 \cdot I'$, *where $\kappa_0 \ge 1$ is arbitrary, Λ is defined in Lemma 6.3 and I' in Lemma 6.4.*

(vi) $\tilde{n} = \max\{1, \lfloor M(\epsilon, l)/\mathrm{cost}(h/2) \rfloor\}$ *for any $\epsilon > 0$, where $h = \kappa_0 \delta_k$ and M and* cost *are defined in* (6.1) *and* (6.2) *below.*

Proof. We follow [3] and only describe the changes. The structure of the argument is that of [3, (4.22)–(4.23)]: if J denotes the number of representatives, then

$$J^2 c_1^2 \le J(c_2 + c_3 + c_4)\frac{X^{2-2\alpha}}{\delta \log T} + J^2 c_5 \quad \text{and} \quad N(\alpha, T) \le \tilde{n} J.$$

Each change below replaces one of Bellotti's constants by a smaller admissible one.

The term B_1 replaces $44.66\, d_2 (v+w)^4 T^{(v+w)/2 + x(1/2 - \alpha_0) + 27/164} \log^4 T$ in [3, (3.17)]. Writing $a = ga'$ and $b = gb'$ with $(a', b') = 1$ and using $|\psi_a|, |\theta_b| \le 1$ and $\sum_{n \le N} n^{-1/2} \le 2\sqrt{N}$, we obtain the bound

$$|M(\tfrac{1}{2} + it)| \le \sum_g g^{-1/2} \cdot 2\Big(\frac{V}{g}\Big)^{1/2} \cdot 2\Big(\frac{W}{g}\Big)^{1/2} < 4\zeta(3/2)\sqrt{VW} < 10.45\sqrt{VW}$$

for the mollifier, which contains no logarithms. Moreover, we use $|\zeta(\frac12+iu)| \leq 0.618|u|^{1/6}\log|u|$ for $3 \leq |u| \leq e^{105}$ by Hiary, Patel and Yang [15], $|\zeta(\frac12+iu)| \leq 1.461$ for $|u| < 3$ [14], and $(\gamma+|t|)^{1/6}\log(\gamma+|t|) \leq T^{1/6}\log T\,(1+|t|)$. This leads to the integral

$$I_\Gamma = \sup_{-1/2\leq\sigma\leq 1/2-\alpha} \int_{\mathbb{R}} |\Gamma(\sigma+it)|(1+|t|)\,dt \leq 5.922 \qquad (\delta \leq 0.03),$$

which we verify in Section 7. The last term of c_1 is Bellotti's bound for the tail $\sum_{n>U}\Psi(n)\Theta(n)n^{-\rho}e^{-nL^2/U}$ [3, (3.19)ff.].

The last factor in c_5 replaces $J_1+J_2+J_3+J_4 \leq 0.241+5.921+286.077T^{27/164}+0.241$ in [3, (4.17)–(4.19)]. For $\delta \leq 0.03$ we have $c := \frac32 - \beta_r - \beta_{r'} \in [-\frac12, -0.44]$. For all $|u| \geq 3$ we have $|\zeta(\frac12+iu)| \leq \max\{1.461, 0.618|u|^{1/6}\log|u|\}$, since for $|u| > e^{105}$ the bound $66.7|u|^{27/164}$ of Patel and Yang [22] is smaller than the right-hand side. Since $|u| \leq T+|t|$, the integral [3, (4.16)] is at most

$$0.618\,T^{1/6}\log T \int_{\mathbb{R}} |\Gamma(c+it)|\Big(1+\frac{|t|}{3}\Big)^{1/6}\Big(1+\frac{\log(1+|t|/3)}{\log 3}\Big)dt \leq 4.72\cdot 0.618\,T^{1/6}\log T;$$

the integral is verified in Section 7 as well.

The quantity d_4 is the Barban–Vehov mean square of [3, Lemma 4.2]. Since Theorem 6.5 bounds the whole sum, Bellotti's condition $u+v<x$ is not needed.

For the off-diagonal terms c_2+c_3, Bellotti assumes in [3, (4.11)] that representatives of distinct rectangles are at least δ apart. This need not be the case, since representatives of adjacent rectangles may be arbitrarily close. Lemma 6.3 below avoids this assumption, and it uses Theorem 5.2. The far term c_4 is treated by Lemma 6.4, and $\tilde n$ replaces $b_1\delta\log T + b_2$ in [3, Lemma 2.4]. □

6.2. **The pair sum.** Let $\mathcal{Z}(\epsilon) = -\zeta'(1+\epsilon)/\zeta(1+\epsilon)$. For $s_0 = 1+\epsilon+i\gamma_r$ with $\gamma_r \leq T$, the Hadamard product and Stirling's formula give

$$\sum_\rho \operatorname{Re}\frac{1}{s_0-\rho} \leq M(\epsilon,L) := \min\Big\{\frac12\log\frac{T}{2\pi}+\mathcal{Z}(\epsilon),\ \min_{1/2\leq a\leq 2}\Big(\frac12\log\frac{T}{2\pi}+\frac{a^2+1/2}{2a}\mathcal{Z}(\epsilon)+\frac{\log(T/\pi)}{8a}\Big)\Big\}, \tag{6.1}$$

where the second expression comes from the non-negativity of $(c+\cos\varphi)^2$. Every term in the sum is positive, and a zero with $\beta \geq 1-\delta$ at vertical distance d from s_0 contributes at least

$$\operatorname{cost}(d) := \min\Big\{\frac{\epsilon}{\epsilon^2+d^2},\ \frac{\epsilon+\delta}{(\epsilon+\delta)^2+d^2}\Big\}, \tag{6.2}$$

since $a \mapsto a/(a^2+d^2)$ is unimodal.

Lemma 6.3. *Fix a representative r and let $h = \kappa_0\delta$. On each side of r, the representative of the j-th rectangle has distance $d \in ((j-1)h, (j+1)h)$ from r. Let $d^* = d^*(\lceil l\rceil, k)$ be as in Theorem 5.2 ($d^* = 0$ outside its range), and let*

$$b_j = \frac{g(m_j)}{m_j}, \qquad m_j = \max\{2\delta, (j-1)h, d^*\}, \qquad g(d) = \min\Big\{1, \frac{|\Gamma(1+2\delta+id)|}{\Gamma(1+2\delta)}\Big\}.$$

Rectangles with $(j+1)h \leq d^*$ contain no representative. Then

$$\sum_{\substack{r'\neq r\\ |\gamma_{r'}-\gamma_r|\leq 1}} |\Gamma(2-\rho_r-\bar\rho_{r'})|X^{2-\beta_r-\beta_{r'}} \leq \frac{1.0067\,X^{2\delta}}{2\delta}\cdot 2\delta\mathcal{L}(h),$$

where $\mathcal{L}(h)$ is the maximum of $\sum_j b_j y_j$ over $0 \le y_j \le 1$ (two variables for each j) subject to $\sum_j \mathrm{cost}((j+1)h) y_j \le M(\epsilon, L) - \mathrm{cost}(0)$, for any $\epsilon > 0$.

Proof. We have $|\Gamma(z)| = |\Gamma(1+z)|/|z|$, and for $0 \le \mathrm{Re}\, z = s \le 2\delta$,

$$|\Gamma(1+s+id)| \le \Gamma(1+s)\frac{|\Gamma(1+2\delta+id)|}{\Gamma(1+2\delta)},$$

since $\prod_n (1 + d^2/(1+s+n)^2)^{-1}$ is increasing in s, and $\Gamma(1+s) \le 1$ for $0 \le s \le 1$. Moreover, X^y/y is increasing for $y \ge 1/\log X$, hence on $[2\nu, 2\delta]$ by the condition $2\nu x L \ge 1$; this gives the factor $1/m_j$, relative to the diagonal term $1.0067 X^{2\delta}/(2\delta)$. Applying (6.1) at $s_0 = 1 + \epsilon + i\gamma_r$ bounds the total cost of the representatives near γ_r, and the zero r itself costs at least $\mathrm{cost}(0)$. Theorem 5.2 applies, since all representatives counted in the cell (k, l) satisfy $1 - \beta \le \delta_k$ and have height at most e^l; for half-integral l we use the integer $\lceil l \rceil$. □

In the computation, $\mathcal{L}(h)$ is bounded from above by the dual value $\mu(M(\epsilon, L) - \mathrm{cost}(0)) + \sum_j \max\{0, b_j - \mu\,\mathrm{cost}((j+1)h)\}$, which is valid for every $\mu \ge 0$.

Lemma 6.4. *Let $\omega > 0$, $\epsilon > 0$ and $t \le T$. Then a window of height ω centred at t contains at most*

$$n(\omega) = \min\Big\{ \Big\lfloor \frac{\frac{1}{2}\log\frac{T}{2\pi} + \mathcal{Z}(\epsilon)}{\min\{f_\omega(\epsilon), f_\omega(\epsilon+\delta)\}} \Big\rfloor,\ \Big\lceil \frac{\omega}{\delta} \Big\rceil + 1 \Big\}, \qquad f_\omega(a) = \frac{a}{a^2 + \omega^2/4},$$

representatives. Consequently, Bellotti's constant c_4 [3, (4.13)–(4.14)] may be computed with

$$I' = \delta\, n(\omega)\Big(\sum_{i \ge 0} g_\alpha(1 + i\omega) \Big), \qquad g_\alpha(y) = e^{y/6} y^{3/2 - 2\alpha} e^{-\pi y/2}, \qquad \omega = \frac{1}{4},$$

in place of $I = \int_1^T g_\alpha(y)\, dy$.

Proof. The first claim follows from (6.1) applied at $s = 1 + \epsilon + it$, as above. For $y \ge 1$ the logarithmic derivative of g_α is $\frac{1}{6} + (\frac{3}{2} - 2\alpha)/y - \frac{\pi}{2} < 0$, so g_α is decreasing and its maximum on the window $[1 + i\omega, 1 + (i+1)\omega]$ is attained at the left endpoint. □

6.3. **The Barban–Vehov mean square.** Let $\Delta = \log(V/U)$, $Y_1 = \log(X/V)$ and $Y_2 = \log(X/U)$, and let γ_E denote Euler's constant.

Theorem 6.5. *Let $\log U \ge 40$, $X > V$, $\tau \in [1.3, 2.5]$ and $\log X/\log V \in [1.1, 2.6]$. Then*

$$\sum_n \frac{\Psi(n)^2}{n} e^{-n/X} \le e^{-1/X} + D_4 + \mathrm{REM} + 2\sqrt{S^\flat S^\sharp} + S^\sharp,$$

where

$$D_4 = \frac{1}{\Delta^2}\Big[\frac{Y_2^2 - Y_1^2}{2} - \gamma_E \Delta + k_1(Y_1 + Y_2 - 2\gamma_E) + 2k_0 \Big], \qquad k_1 \le -0.5853, \quad k_0 \le 0.4214,$$

REM *is the explicit sum of contour remainders given in Lemma A.3, $S^\sharp$ is the explicit bound of Lemma A.5 for the mean square of the non-smooth part of the weights, and $S^\flat = e^{-1/X} + D_4 + \mathrm{REM}$.*

The proof is given in Appendix A. In Lemma 6.2 we take d_4 to be the right-hand side of Theorem 6.5 minus $e^{-L^2/U}$; this is admissible since $\sum_n \Psi(n)^2 n^{-1} e^{-nL^2/U} \ge e^{-L^2/U}$, the term $n = 1$ being $e^{-L^2/U}$. The idea is to replace the weights by a Gaussian smoothing in $\log d$, whose Mellin transform decays like $e^{-\epsilon^2 t^2/2}$; this allows us to close all contours inside a fixed box of height 400, in which $1/\zeta$ is bounded by a rigorous computation, and the main term $e^{-1/X} + D_4$ arises from the residues at $s = 0$, $s = -w$ and $w = 0$.

For example, for the cell $(k, l) = (80, 30)$ of Theorem 6.1, at its parameters, REM ≤ 0.016, $S^\sharp \leq 6.7 \cdot 10^{-5}$ and Theorem 6.5 gives $d_4 \leq 1.62$. Since $\log N^+$ depends on d_4 with elasticity about 1, this is the most important improvement in the density estimate. In the final tables, all cells have $\tau \in [1.54, 1.78]$, $\log X/\log V \in [1.42, 1.55]$ and $\log U \in [56.6, 78.6]$, and all of them use Theorem 6.5.

7. Proof of Theorem 1.1

By Proposition 2.4, it suffices to cover $[43, 3159)$ by intervals $[L_1, L_2]$ and to establish (2.7) on each of them with $q_{[L_1,L_2]} < 1$ whenever $L_2 > 57$. We obtain (2.7) from Proposition 3.5, in which $s_{\max} = 1 - \lambda^*(\log T_1)/\log T_1$ by Theorem 4.6, $S(s)$ is bounded by Lemma 3.6, and $N^+(s,t)$ is the minimum of the bounds (N1)–(N4) of Section 6. For every interval $[L_1, L_2]$ we proceed as follows. First we choose c and a number $f \in [0.3, 0.98]$ by a search over a grid in floating-point arithmetic, and we choose ε by three steps of a fixed-point iteration so that the first term in (3.3) is approximately $f\,\varepsilon_R(L_2+1)/(1+2/L_1)$. Then, with these values of c and ε fixed, we evaluate the right-hand side of (3.3) in ball arithmetic, using the actual value of the first term. The integral over s is bounded by $\sum_j S(s_j)\int_{s_j}^{s_{j+1}} e^{(s-1)L_1}\,ds$, which is admissible since S is non-increasing; the grid consists of 61 points up to $s_{\max} - 0.02$ followed by steps of length $1/(40L_1)$, and its last point is $s_{\max}$ rounded upwards. In Lemma 3.6 we use 80 subintervals of equal length in $\log t$, the last point being $\log T_1$ rounded upwards, and a block is omitted only if the condition of Lemma 3.6 is verified in ball arithmetic. Finally, $q_{[L_1,L_2]}$ is defined as the upper endpoint of the resulting enclosure of $(1+2/L_1)\,|\psi(x)-x|/(x\,\varepsilon_R^-(L_2+1))$, excluding $a(x)/x$.

We start with intervals of width 10, and 2 for $L_1 < 57$, and bisect an interval, down to width 0.5, as long as its value of q (or the floating-point value used in the search) is at least 1. For $L_1 \geq 1100$ the search for (c, f) uses a coarser partition in Lemma 3.6; this affects only the choice of parameters. For $L_1 < 57$ only some bound is needed, since this range enters only through the integral in (2.6). The results are as follows.

range of L	largest value of q	remark
$[43, 57)$	3.85 on $[43, 43.5]$; at most 0.47 on $[53, 57)$	enters only through (2.6)
$[57, 1000)$	0.99137 on $[872, 877]$	width 5 on $[857, 917)$
$[1000, 2000)$	0.8887 on $[1000, 1010]$	
$[2000, 2600)$	0.4064	
$[2600, 3159)$	0.4003	

Hence $q_{[L_1,L_2]} < 1$ for every interval with $L_2 > 57$, and Proposition 2.4 gives (2.5). By Lemma 2.1, this proves the first statement of Theorem 1.1, and the second statement follows as explained in the introduction. □

The binding range $L \approx 900$ corresponds to the bandwidth $T_1 \approx e^{30}$. There the sum over zeros is dominated by hypothetical zeros with $(1-\beta)\log\gamma \in [0.21, 0.32]$ and $\log\gamma \in (28.73, 30]$. On the interval $[900, 910]$ the best kernel gives about 1.002 (roughly 0.40 for the smoothing term and 0.60 for the zeros above H_0), whereas on intervals of width 5 we obtain $q < 1$.

Verification. All computations are performed in ball arithmetic with the Arb library [16] via `python-flint`. Floating-point arithmetic enters only through the choice of parameters: the kernel parameters (c, ε), the parameters (u, τ, w, x) of Lemma 6.2, the auxiliary parameters ϵ, κ_0 and μ in Lemmas 6.3 and 6.4 and in Theorem 6.5, and the points of the partitions. Each of them is fixed as an exact binary number before the bound is evaluated. In detail, the computation consists of the following parts.

(1) The enclosures of ε_R in Lemma 2.2, the constant in Lemma 3.1, and Proposition 2.4.
(2) The constants of Section 4, the iteration in the proof of Theorem 4.6, and the cells of Theorem 5.2.
(3) The inputs of Theorem 6.5; in particular, the supremum of $|\mathcal{A}|$ over the torus of phases for $p \le 100$ is evaluated on a grid with explicit cell enclosures, using interval arithmetic on double precision numbers with outward rounding after every operation.
(4) The 3320 non-empty cells of Theorem 6.1, evaluated from their stored parameters. Here $\mathcal{Z}(\epsilon)$ is computed from the Taylor series of ζ at $1+\epsilon$, $\mathcal{L}(h)$ is bounded by its dual value, and $\tilde{n}$ and $n(\omega)$ are floors of verified upper bounds.
(5) The two integrals in the proof of Lemma 6.2. On $[-\frac{1}{2}, -0.44]\times[0,60]$ we verify $\operatorname{Re}\frac{\Gamma'}{\Gamma}(\sigma+it) > 0$ on boxes, so that $|\Gamma(\sigma+it)|$ is increasing in σ there; the integrals at the right endpoints are computed by rigorous numerical integration, and $|t| > 60$ is estimated explicitly. This gives $I_\Gamma \le 5.92164$ and 4.71363 for the second integral.
(6) The scan described above.

On two processor cores the whole computation takes about 20 hours.

8. Remarks

8.1. **On the literature.** While re-deriving the explicit results which we use, we noticed the following points. None of them affects the main results of the papers concerned.

Remark 8.1. In [4], the term coming from Lemma 10 enters the final inequality as $-0.1186\, w(0)\eta$, that is, with $f(0) = \eta w(0)$. By [4, Definition 2], however, $f(0) = \kappa\eta w(0)$ with $\kappa = 0.504$, so that this term should be $-0.0598\, w(0)\eta$. Evaluating their inequality exactly at the critical height $t = e^{76.47}$ with this correction still gives $\eta \log t \ge 0.2054 > 1/4.896$, so [4, Theorem 1] is not affected. We also note that $F(2\sigma - 1 + \beta_0)$ in [4, (25)] should read $F(\sigma - 1 + \beta_0)$, as in [4, (47)].

Remark 8.2. For $p = 5$, [4, Table 1] gives $m_5 = 0.00735709$. However, $G_5(1, x)$ takes the value $0.0073564919\ldots$ at $x = 1.6495242\ldots$. All other entries of the table are confirmed. The effect on the constant 0.23545 in [4] is -10^{-6}.

Remark 8.3. The majorant $U(y)$ of [4, (31)] integrates to 28.3 against $(x_0^2 + y^2)^{-3/2}$, and not to less than 24 as stated in [4, (32)]. With the digamma function itself the integral is 23.59, so the inequality [4, (32)] is correct. At $(\eta_0, \sigma_0) = (0.0071093, 0.9935164)$, Lemma 4.3 gives $185.4\eta^2 + 7972\eta^3$, compared with $187\eta^2 + 8013\eta^3$ in [4, (48)].

Remark 8.4. The uniform version of [4, (21)] replaces the shifts $(2\sigma - 1)m$ by $c_0 m$ with $c_0 = 151/153$, which requires $\sigma \ge 0.993464$. For $c_0 = 0.9864$, which corresponds to the smaller values of σ used here, their test function takes negative values already for $\varepsilon_0 = 1/2000$; Lemma 4.2 avoids this by working with a fixed σ. Moreover, the numerical verification of $\partial_\sigma G_p < 0$ in [4] is not needed, since it follows from $P \ge 0$, and the omission of the prime powers with $m > 15$ in [4, Lemma 10] is justified by $f \ge 0$.

Remark 8.5. In [6, Proposition 4], quoted as [7, Proposition 1], the main term of the smoothing error is $2\varepsilon x |\nu_c(\alpha)| \log(e^\varepsilon x)/\log B$, where $\nu_c(t) = \int_{-\infty}^t \mu_c$ and $\mu_c(t) = -\int_{-\infty}^t \eta_{c,1}$ for $t < 0$. With $\eta_{c,1}$ normalised as in Section 3, so that $\int \eta_{c,1} = 2$, one has $|\nu_c(0)| \sim \sqrt{2/(\pi c)}$, whereas [7, Remark 1] gives $|\nu_c(0)| \sim (2\pi c)^{-1/2}$, which corresponds to a density of total mass 1. We have not been able to reconcile the two normalisations, and Lemma 3.1 does not depend on this.

Remark 8.6. In [3, (4.11)], representatives of distinct rectangles are assumed to be at least δ apart, which does not seem to be justified; Lemma 6.3 avoids this assumption. Since Bellotti's

other constants are considerably larger than ours, her published bounds are not affected in practice.

8.2. **On two recent preprints.** In [10], it is claimed that (1.1) holds for every $x \geq e^{43.5102147}$. The argument combines $\pi(x) < e\,\pi(x/e)$ with $\pi(x) > x/\log x$ in order to conclude in [10, (29)] that $\pi(x)^2 - \frac{ex}{\log x}\pi(x/e) < 0$. This conclusion requires $\frac{x}{\log x} e\pi(x/e) \geq \pi(x)^2$. However, the first inequality only gives $\frac{x}{\log x} e\pi(x/e) > \frac{x}{\log x}\pi(x)$, and passing from this to $\pi(x)^2$ would require $\pi(x) \leq x/\log x$, which is the reverse of the second inequality and is false for $x \geq 17$. The gap cannot be repaired by sharper inputs of the same kind, since the two sides of (1.1) differ by about $x^2/\log^6 x$, while the bounds used there have relative accuracy of order $(\log x)^{-3/2}$.

In [9], it is claimed that (1.1) holds for every $x \geq e^{547}$. The monotonicity argument used there assumes Sterneck's conjecture $|M(x)| < \frac{1}{2}\sqrt{x}$, which is false [21], and the paper asserts $M(x) = O(\sqrt{x})$, which is an open problem; the proof given yields at most $O(x^{1/2+\varepsilon})$. Moreover, the main term retained there is itself only $O(x^2/\log^2 x)$, so that the sign of $\pi(x)^2 - \frac{ex}{\log x}\pi(x/e)$ is not determined, and the range below $e^{3158.442}$ is treated by evaluating this function at 26 sample points. Hence neither preprint proves (1.1) in the range considered here.

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Appendix A. Proof of Theorem 6.5

Throughout this appendix the notation is local. We fix the numerical parameters

$$\begin{aligned} \varsigma = 0.02, \qquad T_2 = 400, \qquad W_0 = 40, \qquad c = 0.02, \qquad c' = 0.01, \\ \sigma_1 = 0.01, \qquad \theta_1 = 0.40, \qquad \theta_2 = 0.30, \qquad b = 0.38, \qquad \kappa_1 = 0.38, \end{aligned}$$

where ς is the width of a Gaussian smoothing. We write $U = e^{\ell_U}$, $V = e^{\ell_V}$ and $X = e^{\ell_X}$, so that $\Delta = \ell_V - \ell_U$, $Y_1 = \ell_X - \ell_V$ and $Y_2 = \ell_X - \ell_U$, and $E(s) = e^{\varsigma^2 s^2/2}$. We use the following explicit inputs.

(B1) In the box $\mathcal{B} = \{0.6 \leq \sigma \leq 1.3,\ |t| \leq 450\}$ we have $|1/\zeta(s)| \leq M_I := 14.57$, and $|\zeta(s)| \leq M_Z := 11.89$ if moreover $|s-1| \geq 0.1$. These bounds are obtained in ball arithmetic from Taylor models of the entire function $(s-1)\zeta(s)$ on a cover of $\mathcal{B}$ by small discs, with the remainder controlled by Cauchy's estimate and $|\zeta(s)| \leq |s|/|s-1| + |s|/\sigma$. The function $1/\zeta$ is analytic in $\mathcal{B}$, since the Riemann hypothesis holds up to height H_0.

(B2) For $\sigma > 1$ and all t, $|\zeta(\sigma+it)| \leq \zeta(\sigma)$ and $|1/\zeta(\sigma+it)| \leq \zeta(\sigma)/\zeta(2\sigma)$. For $0 < \sigma < 1$ and all t, $|\zeta(\sigma+it)| \leq (1/\sigma + 1/(1-\sigma))(1+|t|)$, which follows from $\zeta(s) = s/(s-1) - s\int_1^\infty \{u\} u^{-s-1}\,du$.

(B3) For $\sigma > -1$ the function $v \mapsto |\Gamma(\sigma+iv)|$ is decreasing on $[0,\infty)$, by the product formula for $|\Gamma|^2$.

A.1. **Mellin representation.** Let $\chi(d) = 1$ for $d \leq U$, $\chi(d) = \log(V/d)/\Delta$ for $U < d \leq V$ and $\chi(d) = 0$ for $d > V$, so that $\xi_d = \mu(d)\chi(d)$ and $\Psi(n) = \sum_{d|n} \mu(d)\chi(d)$. Since

$\frac{1}{2\pi i}\int_{(\sigma)} z^s d^{-s} s^{-2}\,ds = \log^+(z/d)$ for $\sigma > 0$, we have

$$\chi(d) = \frac{1}{2\pi i}\int_{(\sigma)} \hat\chi(s) d^{-s}\,ds, \qquad \hat\chi(s) = \frac{V^s - U^s}{\Delta s^2}.$$

Let Z be a standard normal random variable, and put $\chi^\flat(d) = \mathbb{E}[\chi(de^{-\varsigma Z})]$ and $\chi^\sharp = \chi - \chi^\flat$. Since $\mathbb{E}[e^{\varsigma Z s}] = E(s)$, the function $\chi^\flat$ has the Mellin transform $\hat\chi^\flat(s) = \hat\chi(s)E(s)$, which decays like a Gaussian on vertical lines: $|E(\sigma+it)| = e^{\varsigma^2(\sigma^2-t^2)/2}$. Let $\Psi^\flat(n) = \sum_{d|n}\mu(d)\chi^\flat(d)$, $\Psi^\sharp = \Psi - \Psi^\flat$, and

$$S^\flat = \sum_n \frac{\Psi^\flat(n)^2}{n} e^{-n/X}, \qquad \widetilde{S}^\sharp = \sum_n \frac{\Psi^\sharp(n)^2}{n} e^{-n/X}.$$

By Minkowski's inequality in $\ell^2(e^{-n/X}/n)$,

$$\sum_n \frac{\Psi(n)^2}{n} e^{-n/X} \le \big(\sqrt{S^\flat} + \sqrt{\widetilde{S}^\sharp}\big)^2 = S^\flat + 2\sqrt{S^\flat \widetilde{S}^\sharp} + \widetilde{S}^\sharp. \tag{A.1}$$

Lemma A.1. *For* $\operatorname{Re} w$, $\operatorname{Re}(w+s_1)$, $\operatorname{Re}(w+s_2)$ *and* $\operatorname{Re}(w+s_1+s_2)$ *all positive,*

$$\sum_n \frac{1}{n^{1+w}} \sum_{d_1|n} \frac{\mu(d_1)}{d_1^{s_1}} \sum_{d_2|n} \frac{\mu(d_2)}{d_2^{s_2}} = \frac{\zeta(1+w)\zeta(1+w+s_1+s_2)}{\zeta(1+w+s_1)\zeta(1+w+s_2)}\,\mathcal{A}(w,s_1,s_2),$$

where $\mathcal{A} = \prod_p \mathcal{A}_p$ *with*

$$\mathcal{A}_p = \frac{(1-x_1-x_2+z)(1-z)}{(1-x_1)(1-x_2)}, \qquad \mathcal{A}_p - 1 = \frac{zx_1 + zx_2 - z^2 - x_1x_2}{(1-x_1)(1-x_2)},$$

$x_j = p^{-1-w-s_j}$ *and* $z = p^{-1-w-s_1-s_2}$. *The product converges absolutely if each sum of two of the real parts of* $1+w+s_1$, $1+w+s_2$ *and* $1+w+s_1+s_2$, *and twice the last of them, exceeds* 1, *and* $\mathcal{A}(w,0,s_2) = \mathcal{A}(w,s_1,0) = 1$.

Proof. With $q = p^{-1-w}$, the local factor of the left-hand side is $1 + \frac{q}{1-q}(1-p^{-s_1})(1-p^{-s_2}) = (1-x_1-x_2+z)/(1-q)$. Dividing by the local factor $(1-x_1)(1-x_2)/((1-q)(1-z))$ of the zeta quotient gives $\mathcal{A}_p$. For $s_1 = 0$ we have $x_1 = q$ and $z = x_2$, so $\mathcal{A}_p = 1$, and similarly for $s_2 = 0$. The terms of $\mathcal{A}_p - 1$ are of order $p^{-2-\cdots}$, which gives absolute convergence. □

Since $\chi^\flat(d)$ decays faster than any power of d as $d \to \infty$, Mellin inversion in w (with $e^{-y} = \frac{1}{2\pi i}\int_{(c)} \Gamma(w) y^{-w}\,dw$) and in s_1, s_2 gives

$$S^\flat = \frac{1}{(2\pi i)^3}\int_{(c)}\int_{(\sigma_1)}\int_{(\sigma_1)} \Gamma(w) X^w \hat\chi^\flat(s_1)\hat\chi^\flat(s_2)\, G(w,s_1,s_2)\,ds_2\,ds_1\,dw, \tag{A.2}$$

where G is the right-hand side of Lemma A.1; all integrals converge absolutely.

A.2. **The contour shifts.**

Proposition A.2. *Let* $K_c(w) = \zeta(1+w)H_c(w)$, *where*

$$H_c(w) = \frac{1}{2\pi i}\int_{\mathcal{P}} \frac{\mathcal{A}(w;-w-s,s)\,E(w+s)E(s)}{\zeta(1-s)\zeta(1+w+s)(w+s)^2 s^2}\,ds,$$

and $\mathcal{P}$ *is the path* $-c' - iT_2 \to \kappa_1 - iT_2 \to \kappa_1 + iT_2 \to -c' + iT_2$. *Then* H_c *is analytic for* $-\theta_2 \le \operatorname{Re} w \le c$, $|\operatorname{Im} w| \le W_0$, *and with* $k_1 = H_c(0)$ *and* $k_0 = H_c'(0) + \gamma_E H_c(0)$ *we have*

$$S^\flat \le e^{-1/X} + D_4 + \mathrm{REM},$$

where D_4 *is as in Theorem 6.5 and* REM *is the sum of the eight terms in Lemma A.3.*

Proof. We write (A.2) as $S^\flat = \frac{1}{2\pi i}\int_{(c)} \Gamma(w)X^w F_s(w)\,dw$ with $F_s(w) = \sum_n \Psi^\flat(n)^2 n^{-1-w}$, and split the w-integral into $|\operatorname{Im} w| \leq W_0$ and $|\operatorname{Im} w| > W_0$. On the first part we move the contours in four steps.

Step 1. We move s_2 to $\operatorname{Re} s_2 = -c'$. The only pole crossed is $s_2 = 0$, where $\hat\chi^\flat$ has a simple pole with residue 1; since $G(w, s_1, 0) = 1$ and $\frac{1}{2\pi i}\int \hat\chi^\flat(s_1)\,ds_1 = \chi^\flat(1)$, the residue contributes $\chi^\flat(1)\cdot\frac{1}{2\pi i}\int_{|\operatorname{Im} w|\leq W_0}\Gamma(w)X^w\,dw$. Adding the part $|\operatorname{Im} w| > W_0$ of $S^\flat$, we obtain

$$\chi^\flat(1)e^{-1/X} + \frac{1}{2\pi i}\int_{|\operatorname{Im} w|>W_0}\Gamma(w)X^w\big(F_s(w) - \chi^\flat(1)\big)\,dw.$$

Since $\Psi^\flat(1) = \chi^\flat(1) \in [0,1]$ and $|\Psi^\flat(n)| \leq \tau(n)$, we have $|F_s(w) - \chi^\flat(1)| \leq \frac14 + \sum_{n\geq 2}\tau(n)^2 n^{-1-c} \leq \zeta(1+c)^4/\zeta(2+2c)$, so the second term is at most the first term of R_8 in Lemma A.3. The pole of $\zeta(1+w+s_1+s_2)$ at $s_2 = -w-s_1$ has real part $-c-\sigma_1 < -c'$ and is not crossed.

Step 2. For $|\operatorname{Im} s_1| \leq T_2$ we move s_1 to $\operatorname{Re} s_1 = -\theta_1$, joining the moved segment to the original line $\operatorname{Re} s_1 = \sigma_1$ by horizontal connectors at $\operatorname{Im} s_1 = \pm T_2$. In the region swept, $1+w+s_1 \in \mathcal{B}$, so $1/\zeta(1+w+s_1)$ is analytic. At $s_1 = 0$ the function $\hat\chi^\flat$ has a simple pole with residue 1 and $G(w,0,s_2) = 1$, so this residue equals

$$\frac{1}{2\pi i}\int_{(-c')}\hat\chi^\flat(s_2)\,ds_2\cdot\frac{1}{2\pi i}\int_{|\operatorname{Im} w|\leq W_0}\Gamma(w)X^w\,dw = (\chi^\flat(1)-1)\big(e^{-1/X} - \vartheta\big),$$

where $|\vartheta|$ is at most the first term of R_8; since $\chi^\flat(1) \leq 1$ and $e^{-1/X} - \vartheta > 0$, this residue is non-positive and may be omitted. The pole at $s_1 = -w-s_2$, which is crossed when $|\operatorname{Im}(w+s_2)| \leq T_2$, gives

$$R_x = \frac{1}{(2\pi i)^2}\iint \Gamma(w)X^w\hat\chi^\flat(-w-s_2)\hat\chi^\flat(s_2)\,\frac{\zeta(1+w)\,\mathcal{A}(w;-w-s_2,s_2)}{\zeta(1-s_2)\zeta(1+w+s_2)}\,ds_2\,dw.$$

The integral over the moved segment is R_1, the integrals over the connectors and over the remaining parts of $\operatorname{Re} s_1 = \sigma_1$ make up R_2, and extending R_x to all s_2 costs R_3; the integrand of R_2 has an integrable singularity at $s_1 = -w-s_2$, and Fubini's theorem applies.

Step 3. We expand

$$\Delta^2\hat\chi^\flat(-w-s_2)\hat\chi^\flat(s_2)(w+s_2)^2 s_2^2 = \Big(V^{-w} - V^{-w}\Big(\frac{U}{V}\Big)^{s_2} - U^{-w}\Big(\frac{V}{U}\Big)^{s_2} + U^{-w}\Big)E(w+s_2)E(s_2),$$

and denote by $\Phi(s_2)$ the remaining factor of the integrand. Then Φ has a simple pole at $s_2 = 0$ with residue $-E(w)/w^2$ (from the zero of $1/\zeta(1-s_2)$) and a simple pole at $s_2 = -w$ with residue $E(w)/w^2$ (from the zero of $1/\zeta(1+w+s_2)$), since $\mathcal{A}(w;-w,0) = \mathcal{A}(w;0,-w) = 1$. In each of the four terms we keep the parts of the line $\operatorname{Re} s_2 = -c'$ with $|\operatorname{Im} s_2| > T_2$; together they give R_4. On $|\operatorname{Im} s_2| \leq T_2$ we proceed as follows, always joining the moved segment to the remaining line by horizontal connectors at $\operatorname{Im} s_2 = \pm T_2$. For the two diagonal terms we move the segment to $\mathcal{P}$; this crosses $s_2 = 0$ and gives $(V^{-w}+U^{-w})(K_c(w) + E(w)/w^2)$. In the term with $(U/V)^{s_2}$ we move the segment to $\operatorname{Re} s_2 = b$, crossing $s_2 = 0$; this gives $-V^{-w}E(w)/w^2$, and the moved segment and its connectors give R_5. In the term with $(V/U)^{s_2}$ we move the segment to $\operatorname{Re} s_2 = -b$, crossing $s_2 = -w$; this gives $-U^{-w}(V/U)^{-w}E(w)/w^2 = -V^{-w}E(w)/w^2$, and the moved segment and its connectors give R_6. In all these regions $1-s_2$ and $1+w+s_2$ have imaginary parts of absolute value at most $T_2 + W_0 = 440$, so (B1) applies where the real parts

are at most 1.3. Altogether,

$$R_x = \frac{1}{\Delta^2}\cdot\frac{1}{2\pi i}\int \Gamma(w)X^w\Big[V^{-w}\Big(K_c(w)-\frac{E(w)}{w^2}\Big)+U^{-w}\Big(K_c(w)+\frac{E(w)}{w^2}\Big)\Big]dw \\ + O^*(R_3+R_4+R_5+R_6),$$

where $O^*(R)$ denotes a quantity of absolute value at most R.

Step 4. Finally we move w to $\operatorname{Re} w=-\theta_2$ for $|\operatorname{Im} w|\le W_0$. The integrand has a pole only at $w=0$, since $X^wV^{-w}=e^{Y_1w}$ and $X^wU^{-w}=e^{Y_2w}$. Write $\Gamma(w)e^{Yw}=w^{-1}+(Y-\gamma_E)+(\frac{Y^2}{2}-\gamma_EY+\frac{\gamma_E^2}{2}+\frac{\pi^2}{12})w+O(w^2)$, $K_c(w)=k_1w^{-1}+k_0+O(w)$ and $E(w)/w^2=w^{-2}+\varsigma^2/2+O(w^2)$. The residue of $\Gamma(w)e^{Yw}K_c(w)$ is $k_1(Y-\gamma_E)+k_0$, and that of $\Gamma(w)e^{Yw}E(w)/w^2$ is $\frac{Y^2}{2}-\gamma_EY+\frac{\gamma_E^2}{2}+\frac{\pi^2}{12}+\frac{\varsigma^2}{2}$. Hence the residues at $w=0$ sum to

$$\frac{1}{\Delta^2}\Big[\frac{Y_2^2-Y_1^2}{2}-\gamma_E(Y_2-Y_1)+k_1(Y_1+Y_2-2\gamma_E)+2k_0\Big]=D_4,$$

since $Y_2-Y_1=\Delta$. The integral over $\operatorname{Re} w=-\theta_2$ is R_7, and the connectors at $|\operatorname{Im} w|=W_0$ give the second term of R_8. □

A.3. **The remainders.** For a real number σ and $0\le t_0<t_1\le\infty$ put

$$\mathcal{J}_\nu(\sigma;t_0,t_1)=2\int_{t_0}^{t_1}e^{\varsigma^2(\sigma^2-t^2)/2}\frac{(1+t)^\nu}{\sigma^2+t^2}\,dt,\qquad h(\sigma)=\frac{V^\sigma+U^\sigma}{\Delta},$$

so that $\int_{t_0\le|t|\le t_1}|\hat\chi^\flat(\sigma+it)|(1+|t|)^\nu\,dt\le h(\sigma)\mathcal{J}_\nu(\sigma;t_0,t_1)$. Let $\mathcal{G}_\nu=\int_{|v|\le W_0}|\Gamma(c+iv)|(1+|v|)^\nu\,dv$. For sets I_1,I_2,I_3 of real numbers let $\mathcal{A}^*(I_1,I_2,I_3)$ denote an upper bound for $|\mathcal{A}(w,s_1,s_2)|$ whenever the real parts of $1+w+s_1$, $1+w+s_2$ and $1+w+s_1+s_2$ lie in I_1, I_2 and I_3 (a single number stands for a one-point set). Let $Z_b=\max\{M_Z,1/\sigma_2+1/(1-\sigma_2)\}$ with $\sigma_2=1+c-\theta_1-c'$, $\epsilon_T=e^{-\varsigma^2T_2^2/2}$, $\epsilon_T'=e^{-\varsigma^2(T_2-W_0)^2/2}$, $\zeta_1=\zeta(1+c)$ and $\iota(\sigma)=\zeta(\sigma)/\zeta(2\sigma)$.

Lemma A.3. *Let* $\mathcal{A}_1=\mathcal{A}^*(1+c-\theta_1,1+c-c',\sigma_2)$, $\mathcal{A}_1^\square=\mathcal{A}^*([1+c-\theta_1,1+c+\sigma_1],1+c-c',[\sigma_2,1+c+\sigma_1-c'])$, $\mathcal{A}_x=\mathcal{A}^*(1+c',1+c-c',1)$, $\mathcal{A}_V=\mathcal{A}^*(1-b,1+c+b,1)$ *and*

$\mathcal{A}_U = \mathcal{A}^*(1+b, 1+c-b, 1)$. *Then the remainders in the proof of Proposition A.2 satisfy*

$$
\begin{aligned}
R_1 &\leq \frac{\mathcal{G}_1 e^{c\ell_X}\zeta_1 Z_b M_I\,\iota(1+c-c')\mathcal{A}_1}{(2\pi)^3}\,h(-\theta_1)\mathcal{J}_1(-\theta_1;0,T_2)\,h(-c')\mathcal{J}_1(-c';0,\infty),\\
R_2 &\leq \frac{\zeta_1\iota(1+c-c')\mathcal{A}_1^{\square}}{(2\pi)^3}\Big[2\mathcal{G}_0 e^{c\ell_X} M_I\,\hbar\,\mathcal{Z}\\
&\qquad + \mathcal{G}_1 e^{c\ell_X} h(\sigma_1)\mathcal{J}_1(\sigma_1;T_2,\infty)\,\zeta(1+c+\sigma_1-c')\,\iota(1+c+\sigma_1)\,h(-c')\mathcal{J}_1(-c';0,\infty)\Big],\\
R_3 &\leq \frac{\mathcal{G}_0 e^{c\ell_X}\zeta_1\mathcal{A}_x\iota(1+c')\iota(1+c-c')}{(2\pi)^2}\cdot\frac{(U^{-(c-c')}+V^{-(c-c')})e^{\varsigma^2(c-c')^2/2}\epsilon_T}{\Delta T_2^2}\,h(-c')\mathcal{J}_0(-c';0,\infty),\\
R_4 &\leq \frac{4\mathcal{G}_0 e^{cY_2}\zeta_1\mathcal{A}_x\iota(1+c')\iota(1+c-c')}{(2\pi)^2\Delta^2}\cdot\frac{2e^{\varsigma^2((c-c')^2+c'^2)/2}\epsilon_T}{3(T_2-W_0)^3},\\
R_5 &\leq \frac{\mathcal{G}_0 e^{cY_1}}{\Delta^2}\big(r_V+r_V'\big),\qquad R_6 \leq \frac{\mathcal{G}_0 e^{cY_2}}{\Delta^2}\big(r_U+r_U'\big),\qquad R_7 \leq \frac{(e^{-\theta_2 Y_1}+e^{-\theta_2 Y_2})\,\mathcal{R}}{2\pi\Delta^2},\\
R_8 &\leq \frac{\gamma_0 e^{c\ell_X}\zeta_1^4}{2\pi\zeta(2+2c)}\\
&\qquad + \frac{2\gamma_0(\theta_2+c)e^{cY_2}\mathcal{K}}{\Delta^2},\\
\gamma_0 &= \frac{4\cdot 3.4\,W_0^{1/2}e^{-\pi W_0/2}}{\pi(1-1/(\pi W_0))},
\end{aligned}
$$

where

$$
\begin{aligned}
r_V &= \frac{e^{-\Delta b}\zeta_1\mathcal{A}_V M_I\iota(1+c+b)}{2\pi}\cdot\frac{\pi e^{\varsigma^2((c+b)^2+b^2)/2}}{b^2(b+c)},\\
r_U &= \frac{e^{-\Delta b}\zeta_1\mathcal{A}_U M_I\iota(1+b)}{2\pi}\cdot\frac{\pi e^{\varsigma^2((c-b)^2+b^2)/2}}{b^2(b-c)},\\
r_V' &= \frac{2(b+c')e^{c'\Delta}e^{\varsigma^2((c+b)^2+b^2)/2}\epsilon_T\epsilon_T'}{2\pi T_2^2(T_2-W_0)^2}\\
&\qquad\times\zeta_1\,\mathcal{A}^*([1-b,1+c'],[1+c-c',1+c+b],1)\,M_I\,\iota(1+c-c'),\\
r_U' &= \frac{2(b-c')e^{\varsigma^2 b^2}\epsilon_T\epsilon_T'}{2\pi T_2^2(T_2-W_0)^2}\\
&\qquad\times\zeta_1\,\mathcal{A}^*([1+c',1+b],[1+c-b,1+c-c'],1)\,\iota(1+c')\,M_I.
\end{aligned}
$$

Here $\hbar = (V^{\sigma_1}+U^{\sigma_1})e^{\varsigma^2(\theta_1^2-T_2^2)/2}/(\Delta T_2^2)$ *bounds* $|\hat{\chi}^\flat|$ *on the connectors of Step 2,*

$$
\begin{aligned}
\mathcal{Z} = (\theta_1+\sigma_1)\Big(\Big(2+\frac{2+W_0+T_2}{\sigma_2}\Big)h(-c')\mathcal{J}_0(-c';0,\infty)+\frac{h(-c')\mathcal{J}_1(-c';0,\infty)}{\sigma_2}\Big)\\
+2\pi\sqrt{(\theta_1+\sigma_1)^2+1}\,\frac{(V^{-c'}+U^{-c'})e^{\varsigma^2 c'^2/2}}{\Delta c'^2},
\end{aligned}
$$

$\mathcal{R} = 2\int_0^{W_0} |\Gamma(-\theta_2+iv)|\big(\mathcal{K}(v)+e^{\varsigma^2\theta_2^2/2}\theta_2^{-2}\big)\,dv$, $\mathcal{K} = \mathcal{K}(W_0)+e^{\varsigma^2\theta_2^2/2}W_0^{-2}$, *and*

$$\mathcal{K}(v) = \frac{M_Z}{2\pi}\Big(\mathcal{A}^*(1-\kappa_1,[1+\kappa_1-\theta_2,1+\kappa_1+c],1)M_I\,\iota(1+\kappa_1-\theta_2)\frac{\pi(2\kappa_1-\theta_2)e^{\varsigma^2((\kappa_1+c)^2+\kappa_1^2)/2}}{\kappa_1(\kappa_1-\theta_2)((2\kappa_1-\theta_2)^2+v^2)}$$
$$+\,\mathcal{A}^*([1-\kappa_1,1+c'],[1-\theta_2-c',1+c+\kappa_1],1)M_I^2\,\frac{2(\kappa_1+c')e^{\varsigma^2((\kappa_1+c)^2+\kappa_1^2)/2}\epsilon_T\epsilon_T'}{T_2^2(T_2-W_0)^2}\Big).$$

The function $\mathcal{K}(v)$ *bounds* $|K_c(w)|$ *for* $\operatorname{Re} w=-\theta_2$ *and* $|\operatorname{Im} w|=v$, *and also for* $-\theta_2\le\operatorname{Re} w\le c$ *and* $|\operatorname{Im} w|=W_0$.

Proof. Each term is the integral of the absolute value of the integrand over the corresponding piece of contour, bounded factor by factor; several of the resulting bounds are weaker than necessary by small constant factors. The factor $\Gamma(w)X^w$ gives $\mathcal{G}_\nu e^{c\ell_X}$, or $\mathcal{G}_\nu e^{cY_j}$ after V^{-w} or U^{-w} has been absorbed; in R_4, all four coefficients are at most e^{cY_2}, since $e^{cY_1+c'\Delta}\le e^{cY_2}$. The factors $\hat\chi^\flat$ give $h(\sigma)\mathcal{J}_\nu(\sigma;\cdot,\cdot)$. The zeta factors are bounded by (B1) when the argument lies in $\mathcal{B}$, and by (B2) otherwise; for real parts in $[1.3,1.4]$ we use $\iota(1.3)<M_I$. In R_1 we use $|\zeta(1+w+s_1+s_2)|\le Z_b(1+|\operatorname{Im} w|)(1+|\operatorname{Im} s_1|)(1+|\operatorname{Im} s_2|)$ on $\operatorname{Re}(1+w+s_1+s_2)=\sigma_2$, which follows from (B1) for heights at most 450 and from (B2) beyond; the three factors $1+|\operatorname{Im}|$ are absorbed into $\mathcal{G}_1$ and $\mathcal{J}_1$. In R_2, on the connectors, the pole of $\zeta(1+u)$ with $u=w+s_1+s_2$ is handled by $|\zeta(1+u)|\le 1/|u|+1+|1+u|/\operatorname{Re}(1+u)$, integrating $1/|u|$ over a disc of radius $\sqrt{(\theta_1+\sigma_1)^2+1}$ and using $1/|u|\le 1$ elsewhere. Whenever a contour runs at height at least T_2, the factor $|E(s)|$ gives ϵ_T and, where it is present, $|E(w+s)|$ gives ϵ_T', while $|s|\ge T_2$ and $|w+s|\ge T_2-W_0$. The identity $\int_{\mathbb{R}} dt/((t^2+a^2)((t+v)^2+a'^2))=\pi(a+a')/(aa'((a+a')^2+v^2))$ gives the factors in r_U, r_V and $\mathcal{K}(v)$. In $\mathcal{K}(v)$, $|\zeta(1+w)|\le M_Z$ holds since $|w|\ge\theta_2$ on $\operatorname{Re} w=-\theta_2$ and $|w|\ge W_0$ on the connectors. In R_8 we use the bound for $|F_s(w)-\chi^\flat(1)|$ from Step 1 and, for $v\ge W_0$ and $-\theta_2\le\sigma\le c$,

$$|\Gamma(\sigma+iv)| = \frac{|\Gamma(\sigma+1+iv)|}{|\sigma+iv|}\le\Gamma(\sigma+1)\frac{|\Gamma(2+iv)|}{v}$$
$$=\Gamma(\sigma+1)\frac{\sqrt{1+v^2}}{v}\sqrt{\frac{\pi v}{\sinh\pi v}}\le 3.4\,v^{1/2}e^{-\pi v/2},$$

where we used that $|\Gamma(x+iv)|/\Gamma(x)$ is increasing in $x>0$ and $\Gamma(\sigma+1)\le\Gamma(0.7)<1.3$. □

All quantities in Lemma A.3 which do not depend on (U,V,X) (the integrals $\mathcal{G}_\nu$, $\mathcal{J}_\nu$ and $\mathcal{R}$, the values of ζ, and the bounds $\mathcal{A}^*$) are computed once in ball arithmetic. The integrals are upper Riemann sums, using (B3) and the monotonicity of the other factors. For $\mathcal{A}^*$ we bound $\prod_{p\le100}|\mathcal{A}_p|$ by the product of the suprema of $|\mathcal{A}_p|$ over the 3-torus of phases of x_1, x_2 and z, evaluated on a grid of 96^3 cells with an explicit cell enclosure (moving a phase by at most $h/2$ moves x_j by at most $|x_j|h/2$ and z by at most $3|z|h/2$), and with the moduli of x_1, x_2, z given as intervals when the real parts vary. Where the real parts vary along a contour, the range is cut into up to 32 pieces, according to the real parts of the moving variables, and the maximum over the pieces is taken. For $100<p\le10^6$ we use $\log|\mathcal{A}_p|\le|\mathcal{A}_p-1|$ and the triangle inequality, and for $p>10^6$ the bound $4.2\cdot10^{6(1-m)}/(m-1)$, where m is the least of the four sums of pairs of real parts in Lemma A.1. For example, $\mathcal{A}_1\le1202.27$ and $\mathcal{A}_x\le14.56$.

A.4. **The non-smooth part.**

Lemma A.4. *Let $\varpi(y) = \phi(|y|) - |y|\bar\Phi(|y|)$, where ϕ and $\bar\Phi$ are the standard normal density and tail, and $\Phi_1(y) = \int_y^\infty \varpi$. For real A put $\Omega(A) = \sum_{r\geq 1} \varpi((\log r - A)/\varsigma)/r$. Then*

$$\Omega(A) \leq \varsigma\Big(\frac{1}{2} - \Phi_1\Big(\frac{A}{\varsigma}\Big)\Big) + 0.8\, e^{\varsigma^2/2} e^{-A} \quad (A \geq 0), \qquad \Omega(A) \leq \varsigma\Phi_1\Big(\frac{|A|}{\varsigma}\Big) + 2\varpi\Big(\frac{|A|}{\varsigma}\Big) \quad (A < 0).$$

In particular $\Omega(A) \leq \varsigma/2 + 0.8e^{\varsigma^2/2}$ for all A, and $\Omega(A) \leq \varsigma/2 + 0.8e^{\varsigma^2/2}e^{-A}$ for $A \geq 0$.

Proof. The function ϖ is even and log-concave, since $\varpi(y) = \int_{|y|}^\infty \bar\Phi$ for $y \geq 0$ and $\bar\Phi$ is log-concave. Hence $u \mapsto \varpi((u-A)/\varsigma)e^{-u}$ is log-concave, and $g(t) = \varpi((\log t - A)/\varsigma)/t$ is unimodal on $[1,\infty)$. For a unimodal function, $\sum_{r\geq 1} g(r) \leq \int_1^\infty g + 2\max g$. Here $\int_1^\infty g(t)\,dt = \varsigma\int_{-A/\varsigma}^\infty \varpi = \varsigma\Phi_1(-A/\varsigma)$, and $\Phi_1(-y) = \frac{1}{2} - \Phi_1(y)$ since $\int_{\mathbb{R}} \varpi = \frac{1}{2}$. Since $\varpi \leq \phi$ and $\phi(y)e^{-\varsigma y} = \phi(y+\varsigma)e^{\varsigma^2/2}$, we have $\max g \leq e^{-A}\phi(0)e^{\varsigma^2/2} \leq 0.4e^{\varsigma^2/2}e^{-A}$. For $A < 0$ the function g is decreasing on $[1,\infty)$, so $\sum_r g(r) \leq g(1) + \int_1^\infty g = \varpi(|A|/\varsigma) + \varsigma\Phi_1(|A|/\varsigma)$. □

Lemma A.5. *We have $|\chi^\sharp(d)| \leq \frac{\varsigma}{\Delta}[\varpi(\frac{\log d - \ell_U}{\varsigma}) + \varpi(\frac{\log d - \ell_V}{\varsigma})]$ for every $d \geq 1$. Let $\beta(m) = \Omega(\ell_U - \log m) + \Omega(\ell_V - \log m)$, and let Σ be an upper bound for $\sum_m \mu^2(m)\varphi(m)^{-1}\beta(m)^2$. Then, for every $\epsilon > 0$,*

$$\widetilde{S}^\sharp \leq S^\sharp := \Big(\frac{\epsilon}{e}\Big)^\epsilon X^\epsilon\Big(\frac{1}{\epsilon} + 1\Big)\Big(\frac{\varsigma}{\Delta}\Big)^2\Sigma + \bar\Phi(\ell_U/\varsigma)^2,$$

and an admissible value of Σ is

$$\Sigma = \beta_0^2\Big(\log 10^6 + 1.3326 + \frac{4}{1000}\Big) + \sum_j \beta_j^2\Big(\delta_u + 8e^{-u_j/2}\Big) + \frac{36(\varsigma+2)^2 e^{-y_0^2}}{2\pi}\Big(1 + \frac{\varsigma}{2y_0}\Big),$$

where $\beta_0 = \varsigma + 0.8e^{\varsigma^2/2}(e^{-(\ell_U - \log 10^6)} + e^{-(\ell_V - \log 10^6)})$, the $u_j = \log 10^6 + j\delta_u$ with $\delta_u = 0.02$ run over $[\log 10^6, \ell_V + 60\varsigma)$, β_j is the upper bound for $\beta(m)$ on $\log m \in [u_j, u_j + \delta_u]$ given by Lemma A.4 (using the monotonicity of the bounds in A), and $y_0 = 60$.

Proof. As a function of $y = \log d$, χ is piecewise linear with a change of slope of $-1/\Delta$ at ℓ_U and of $+1/\Delta$ at ℓ_V. Since $\mathbb{E}[(y - \varsigma Z)^+] - y^+ = \varsigma\varpi(y/\varsigma)$, this gives the bound for $\chi^\sharp$. Next, $e^{-y} \leq (\epsilon/e)^\epsilon y^{-\epsilon}$ for $y > 0$, and for any f,

$$\sum_n \frac{1}{n^{1+\epsilon}}\Big(\sum_{d|n}\mu(d)f(d)\Big)^2 = \zeta(1+\epsilon)\sum_m \frac{\mu^2(m)}{m^{1+\epsilon}}\prod_{p|m}\Big(1 - \frac{1}{p^{1+\epsilon}}\Big)\Big(\sum_r \frac{\mu(r)f(mr)}{r^{1+\epsilon}}\Big)^2,$$

which follows from $\gcd(d_1,d_2)^{1+\epsilon} = \sum_{m|\gcd(d_1,d_2)} m^{1+\epsilon}\prod_{p|m}(1-p^{-1-\epsilon})$. Since $p^{-1-\epsilon}(1 - p^{-1-\epsilon}) \leq 1/(p-1)$ and $\zeta(1+\epsilon) \leq 1/\epsilon + 1$, and $\sum_r |\chi^\sharp(mr)|/r \leq \frac{\varsigma}{\Delta}\beta(m)$, we obtain the bound for $\widetilde{S}^\sharp$; the term $\bar\Phi(\ell_U/\varsigma)^2$ is not needed but is kept, as in the computation. For Σ we split the sum over m. For $m \leq 10^6$ we have $\ell_K - \log m \geq 26$ for $K \in \{U, V\}$, so $\beta(m) \leq \beta_0$ by Lemma A.4, and $\sum_{m\leq y}\mu^2(m)/\varphi(m) = \log y + c_0 + O^*(4/\sqrt{y})$ with $c_0 = 1.33258\ldots < 1.3326$ [25]. For $\log m \in [u_j, u_j + \delta_u]$ the same estimate shows that the total mass of $\mu^2(m)/\varphi(m)$ is at most $\delta_u + 8e^{-u_j/2}$. For $\log m \geq \ell_V + 60\varsigma$ both arguments of b are at most -60ς, so $\beta(m) \leq 2(\varsigma+2)\phi((\log m - \ell_V)/\varsigma)$, and the mass on intervals of length 1 is at most 9; comparing the sum over such intervals with an integral of the decreasing majorant gives the last term. □

A.5. **Proof of Theorem 6.5.** By (A.1), Proposition A.2 and Lemma A.5, the sum in Theorem 6.5 is at most $S^\flat + 2\sqrt{S^\flat S^\sharp} + S^\sharp$ with $S^\flat \le e^{-1/X} + D_4 + \mathrm{REM}$. It remains to bound $k_1 = H_c(0)$ and $k_0 = H_c'(0) + \gamma_E H_c(0)$. At $w = 0$ the integrand of H_c is $g(s) = \mathcal{A}(0; -s, s) e^{\varsigma^2 s^2}/(\zeta(1-s)\zeta(1+s)s^4)$, and differentiation with respect to w multiplies it by $D(s) = \frac{\partial}{\partial w}\log\mathcal{A} + \varsigma^2 s - \frac{\zeta'}{\zeta}(1+s) - \frac{2}{s}$. On the segment $\operatorname{Re} s = \kappa_1$, $|\operatorname{Im} s| \le 150$, we compute $\frac{1}{2\pi i}\int g$ and $\frac{1}{2\pi i}\int gD$ by rigorous numerical integration, with the Euler product truncated at $p \le 2000$. The error from the truncation is at most $(e^\tau - 1)\int |g|$, with τ obtained from $|\mathcal{A}_p - 1| \le 5.2p^{-m}$ ($m = 2 - \kappa_1$) and the bound $\pi(t) \le 1.26t/\log t$ of Rosser and Schoenfeld [26], and similarly for D; here $\int |g| \le (\pi \int |g|^2(1+t^2)\,dt)^{1/2}$ by the Cauchy–Schwarz inequality. On the parts $150 < |\operatorname{Im} s| \le T_2$ and on the two horizontal pieces of $\mathcal{P}$ we bound $|g|$ and $|gD|$ in ball arithmetic on pieces of length 0.02, using the truncated product and the tail bounds $|\mathcal{A}/\mathcal{A}_P - 1| \le e^{\tau'} - 1$ with $\tau' = 4.4P^{1-m'}/(m'-1)$ and $m' = 1.62$. This gives $k_1 \in -0.607063 \pm 0.0217$ and $k_0 \in 0.35026 \pm 0.0711$, hence $k_1 \le -0.5853$ and $k_0 \le 0.4214$. Finally, the hypothesis $X > V$ gives $Y_1 > 0$, which is used in Step 4, and $\ell_U \ge 40$ is used in Lemma A.5; the conditions on τ and $\log X/\log V$ are those under which the computation was carried out. □

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